How fast is fast enough? The epidemiological benefit of shortening the symptom-onset-to-isolation delay in the 2026 Bundibugyo virus disease outbreak, Democratic Republic of the Congo: a transmission-modelling study

Team TAUS, MPPR 2026 group project. Advisers: Dr James Azam, Andrej Jarynowski. Correspondence: olagbami.samfem@gmail.com. Preliminary results, November 2026

Oluwafemi Samson Olagbami Humphrey Rop Imelda Laetitia Simboro
Jonathan Boakye-Yiadom

Abstract

Background. The 2026 outbreak of Bundibugyo virus disease (BVD) in the Democratic Republic of the Congo (DRC) has exceeded 10 000 confirmed cases. Every day between symptom onset and isolation is a day of possible onward transmission, yet the value of shortening that delay when isolation is imperfect has not been quantified for this outbreak. **Methods.** We built a deterministic susceptible–exposed–infectious–isolated–removed (SEIQR) model in which the isolation rate is the inverse of the mean onset-to-isolation delay and isolated people transmit at a fraction $1 - \epsilon$ of the rate of non-isolated people. We fitted it by maximum likelihood to national surveillance data up to 12 September 2026 (confirmed cases by onset date, corrected for reporting delay, and by report date), allowing one change in transmission. We simulated nine scenarios (delays of 7, 5 and 3 days $\times$ isolation effectiveness ϵ of 50%, 80% and 100%) from 12 September, propagating fitted and literature uncertainty through 1,000 paired parameter draws, a stochastic version of the model, global and structural sensitivity analyses, and an out-of-sample forecast check. **Findings.** Transmission grew ($R_e \approx 1.20$) until about 24 July and then plateaued ($R_e \approx 1.00$); with a 7-day delay the model projected 31,089 (9,845 to 107,831) symptomatic cases over 12 months (median and 95% uncertainty interval [UI]). Reducing the mean delay to 5 days averted 50%, 73% and 84% of cases at 50%, 80% and 100% isolation effectiveness; reducing it to 3 days averted 75%, 89% and 94%. At 80% effectiveness, R_e fell from 1.00 to 0.89 (5 days) and 0.74 (3 days), and the median time to the end of transmission was 24 and 9 months, against more than 60 months at 7 days. Delay and isolation effectiveness were the dominant uncertainties (partial rank correlations 0.94 and -0.80). The benefit of a 5-day target was reversed if the true current delay were already below 5 days. A forecast made with data to 12 September overestimated the next 46 days of cases by 23%. **Interpretation.** Faster isolation can turn a plateau into decline, but quality of isolation matters as much as speed, and the size of the effect rests on the outbreak currently sitting at the epidemic threshold. Measuring the actual onset-to-isolation delay should

be a surveillance priority. **Funding.** None declared (to be confirmed by the authors).
Keywords: Bundibugyo virus; Ebola disease; isolation; transmission model; scenario analysis; uncertainty; reproducibility

Key points

- **Question:** how many cases would be prevented, and how much sooner would transmission end, if the mean delay from symptom onset to isolation in the DRC fell from 7 to 5 or 3 days, allowing for isolation that is only 50–100% effective?
- **Answer:** about 73% (5 days) and 89% (3 days) of the next 12 months' cases at 80% effectiveness; 75% even at 50% effectiveness with a 3-day delay.
- **Caveats:** effects are large because the outbreak is near the threshold $R_e \approx 1$; the baseline delay is poorly measured; absolute counts are conditional on a plateau that later eased.

1 Introduction

In May 2026 the DRC declared an outbreak of Ebola disease caused by Bundibugyo virus, which has become the second-largest Ebola-disease outbreak on record, with a crude case fatality of about 48% among confirmed cases [Zomahoun et al., 2026, Kabasele et al., 2026]. By 28 October 2026, 10 274 cases and 4 992 deaths had been confirmed in the DRC [US Centers for Disease Control and Prevention, 2026]. Control relies on finding cases, isolating them, and tracing their contacts. Operational reviews in August found that all contact-tracing indicators were below target, that only 15–20% of new cases were known contacts, that treatment-unit occupancy averaged 64% (with some facilities above 100%), and that 59% of confirmed deaths occurred outside treatment units [Kabasele et al., 2026]. The delay between symptom onset and isolation is therefore a central operational quantity: community engagement, decentralised testing, rapid result return and transit centres all aim to shorten it. In the first month of the outbreak the mean delay from first symptoms to sampling was reported as 7.4 days, and a preprint from the response gives 4.6 days from onset to sampling but 8.6 days to the test result [Ebengo et al., 2026].

Field teams cannot measure what a shorter delay would buy; a model can. Theory and experience from past Ebola outbreaks show that the combination of how quickly cases are isolated and how completely isolation prevents transmission determines whether an outbreak is controllable [Fraser et al., 2004, Chowell and Nishiura, 2014, Pandey et al., 2014, Kucharski et al., 2015]. Scenario projections for the present outbreak have varied the *fraction* of symptomatic people isolated while holding the delay fixed at two days and assuming that isolated people transmit nothing [Mooring et al., 2026]. To our knowledge, the benefit of shortening the delay itself, when isolation is imperfect and when the model is anchored to the observed course of this outbreak, has not been reported.

Research question. Among symptomatic infectious people in the DRC outbreak, how would reducing the mean onset-to-isolation delay from 7 days to 5 and 3 days change cumulative cases and outbreak duration, under worst-, base- and best-case isolation effectiveness (50%, 80%, 100%)? We answer it with a transmission model fitted to surveillance data and subjected to explicit tests of its own weaknesses: reporting delay, structural assumptions about the delay distribution, an out-of-sample forecast, and a stochastic version. The analysis is fully scripted in R so that every figure, table and number in this paper can be regenerated from the data folder.

2 Methods

2.1 Design, data and ethics

This is a modelling study of publicly available, aggregated surveillance data; no individual-level data were used and no ethical approval was required (to be confirmed by the authors' institution). The scenario analysis is counterfactual: because the surveillance data end on **12 September 2026**, scenarios are applied from that date. Later official cumulative counts (27 September and

28 October) were withheld from fitting and used only for validation.

Data are the cleaned national surveillance streams compiled from the daily situation reports (SitReps) of the DRC Institut National de Santé Publique (INSP) and INRB transcriptions by the BVDOutbreakSize project [Abbott et al., 2026], as of 12 September 2026 (7,200 confirmed cases and 3,475 confirmed deaths). Table 1 lists what was used and what was not. The README of the data set warns of changing case definitions, retrospective harmonisation, a digitised onset curve and right-truncation of recent onset dates; we address each below.

Table 1: Data used. Streams not listed (suspected cases, bed capacity, treatment-centre movements) stop in July or August or change definition, and suspected and confirmed cases must not be mixed in one likelihood.

Stream	Period	Use
Confirmed cases by symptom-onset date (SitRep 121; 55 earlier vintages)	25 May–6 Sep (105 days)	**Fit A**. Digitised bar chart (about 66% of confirmed cases have a recorded onset date). The 55 vintages give the reporting-delay curve that corrects the incomplete last three weeks.
Cumulative confirmed cases (by report date)	2 Jun–12 Sep (96 report intervals)	**Fit B (primary)**. Increments between report dates. The 22 July harmonisation step (369 reported, 97 new) is replaced by the printed 97, removing 272 retrospective cases.
Samples analysed per day	1 Jun–12 Sep	Diagnostic only: positivity (new confirmed / samples analysed).
Confirmed deaths (cumulative)	14 May–12 Sep	Crude fatality ratio for an illustrative deaths-averted figure.
Official cumulative confirmed cases, 27 Sep and 28 Oct	–	Out-of-sample validation only [UK Health Security Agency, 2026, US Centers for Disease Control and Prevention, 2026].

2.2 Model

We use a deterministic compartmental model with susceptible (S), exposed (E), infectious but not yet isolated (I), isolated infectious (Q) and removed (R) people (Figure 1):

$$\begin{aligned} \dot{S} &= -\lambda S, & \dot{E} &= \lambda S - \sigma E, & \dot{I} &= \sigma E - (\tau + \gamma) I, \\ \dot{Q} &= \tau I - \gamma Q, & \dot{R} &= \gamma (I + Q), & \lambda &= \beta \frac{I + (1 - \epsilon) Q}{N}, \end{aligned} \tag{1}$$

where σ^{-1} is the mean latent period, γ^{-1} the mean infectious period (the same for isolated people), β the transmission rate, N the population at risk and $\epsilon \in [0, 1]$ the *isolation effectiveness*: isolated people transmit at a fraction $1 - \epsilon$ of the rate of non-isolated people. The intervention is the isolation rate $\tau = 1/\text{delay}$, so the mean delay studied is the mean time to isolation of a person who

remains infectious and not removed. Symptomatic cases accumulate as $\dot{C} = \sigma E$. With $S \approx N$ the reproduction number is

$$R_e = \frac{\beta}{\gamma}\left(1 - \epsilon\,\frac{\tau}{\tau+\gamma}\right), \tag{2}$$

and the share of cases isolated before recovery or death is $\tau/(\tau+\gamma)$ (Appendix A). With a 7-day delay and a 7-day infectious period this share is 50%; it is 58% at 5 days and 70% at 3 days. This competing-risk property of an exponential delay matters for interpretation and is tested by the structural analyses below.

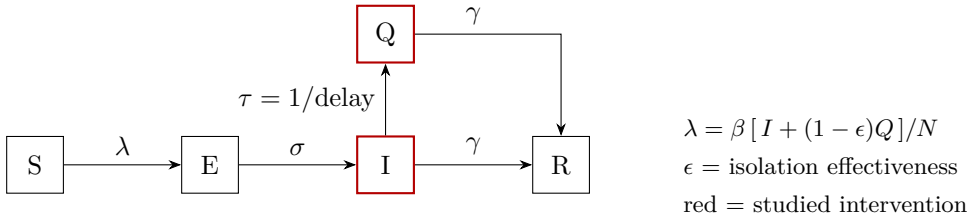


Figure 1: Model structure. The studied intervention is the flow from I to Q at rate $\tau = 1/\text{delay}$; isolation is imperfect when $\epsilon < 1$.

2.3 Parameters

Table 2 gives all inputs. Natural-history values come from the model inputs of the response preprint [Ebengo et al., 2026] and the Ebola review of Chowell and Nishiura [Chowell and Nishiura, 2014]; ranges are triangular distributions that span published estimates (latent period 4.0–10.1 days, mean 6.3; infectious period 5–10 days, mean 7). We did not use the systematic-review database of Nash et al. [Nash et al., 2024], which should replace these ranges in future work. The baseline delay of 7 days is the value in the study brief (first-month mean 7.4 days); we could not locate that figure in the primary source available to us, so two alternative baselines (4.6 and 8.6 days, the preprint's onset-to-sample and onset-to-result means) are analysed as sensitivity cases.

Table 2: Parameters and their sources. UI: uncertainty interval.

Parameter	Value	Source / treatment
Latent period $1/\sigma$	triangular(4.0, 6.3, 10.1) d	[Ebengo et al., 2026, Chowell and Nishiura, 2014]; fixed at 6.3 d in fitting, varied in scenarios
Infectious period $1/\gamma$	triangular(5, 7, 10) d	[Ebengo et al., 2026, Chowell and Nishiura, 2014]; fixed at 7 d in fitting, varied in scenarios
Baseline delay $1/\tau_0$	7 d	Study brief; 4.6 and 8.6 d in sensitivity analyses
Scenario delays	7, 5, 3 d	Study design
Isolation effectiveness ϵ	0.5, 0.8, 1.0	Worst, base, best case
Transmission β	estimated	Fitted (Fit B) or calibrated to the fitted growth rate in each draw
Starting incidence, growth rate, change date	estimated	Maximum likelihood, profile likelihood and Hessian draws
Count dispersion k	estimated	Negative-binomial likelihood
Population N	4 392 200	Ituri province (scalars file); depletion negligible, 10 million tested
Crude fatality ratio	48%	3 475 / 7 200 confirmed at 12 Sep; an upper bound, used only for illustration
Horizon	12 months (cases); 60 months (duration)	Study design

2.4 Fitting

Reporting delay. The onset curve is published in every SitRep, so each onset date's count is observed repeatedly as reports accumulate. We measured the share $c(L)$ of an onset day's final count already visible L days later, pooling the 55 vintages and taking counts at least 28 days old in the latest vintage as final. A Weibull cumulative distribution, $c(L) = 1 - \exp\{-((L+0.5)/s)^{\kappa}\}$, was fitted by least squares. Half of cases were visible after 7 days and 90% after 15 days (mean 8.5 days).

Observation model and likelihood. Counts y are negative-binomial with mean μ and variance $\mu + \mu^2/k$. In Fit A (onset curve) $\mu_d = c(L_d)\, i(d)$, where $i(d)$ is the model's symptomatic incidence on day d and L_d the days between d and the report date of 12 September. In Fit B (report date) the expected increment over a report interval is the sum, over its days u, of $\sum_{d \le u} i(d)\{c(u-d) - c(u-d-1)\}$, i.e. incidence convolved with the same delay. Fit B is the primary fit because it covers all confirmed cases and is not affected by truncation; Fit A is the cross-check.

Parameters estimated. Natural-history parameters and the baseline delay are fixed at their central values; we estimate the starting incidence i_0, the transmission rate before (β_1) and after

(β_2) a change on date t_c, and k. We compared this with constant transmission by AIC (counting t_c as a parameter). Optimisation used Nelder–Mead from multiple stages; the date t_c was profiled on a grid (5-day steps, refined to 1 day). Uncertainty draws sample t_c with weights proportional to the profile likelihood and the other parameters from the multivariate normal given by the Hessian at that t_c. Model adequacy was checked by 95% predictive coverage and residual autocorrelation, and by refitting at extreme natural-history values.

2.5 Scenarios, outcomes and uncertainty propagation

Each of 1,000 draws contains the fitted post-change growth rate r and incidence on 12 September $i(T)$ (jointly, from Fit B), and latent and infectious periods drawn independently from their triangular distributions. For each draw and each ϵ, β is solved so that the 7-day baseline has growth rate r (the dominant eigenvalue of the linearised (E, I, Q) system), the initial state is the corresponding eigenvector scaled to $i(T)$, and the three delays are then simulated by changing only τ. All nine scenarios share the same draws, so cases averted are within-draw differences from the 7-day run with the same ϵ. This choice means that β, and therefore ϵ, are not separately identifiable from case counts: ϵ is a scenario assumption.

Outcomes. *Primary:* cumulative symptomatic confirmed cases over the next 12 months, cases averted (absolute and percentage), and outbreak duration, defined as the first day with fewer than one exposed or infectious person within 60 months (otherwise right-censored; we report medians with censoring, and the probability of ending by 12, 24 and 60 months). *Secondary:* R_e, probability $R_e < 1$, peak incidence and time to peak. Deaths averted are illustrative only (cases averted $\times$ 48%). Results are medians with 95% UIs (2.5th–97.5th percentiles). A policy surface (delays of 2–10 days, ϵ of 30–100%; 100 draws) gives the continuous picture.

2.6 Stochastic model

A discrete-time binomial-chain version ($\Delta t = 0.25$ d, same transitions as Eq. 1) was run for 500 replicates per scenario, each with its own parameter draw. Transmission was calibrated in discrete time so that the expected growth rate equals the deterministic one (an uncorrected calibration biased results upwards in a preliminary run).

2.7 Sensitivity and structural analyses

Global: partial rank correlation coefficients (PRCC) [Blower and Dowlatabadi, 1994, Marino et al., 2008] for 12-month cases over 1,500 samples of latent period, infectious period, growth rate, starting incidence, $\epsilon \sim U(0.5, 1)$ and delay $\sim U(3, 7)$ days, with 300 bootstrap resamples for intervals. **Variants** (17 analyses at $\epsilon = 0.8$, 300 draws each): extreme natural-history values; growth rate and starting incidence at their 2.5th and 97.5th percentiles; fitting to the onset curve instead; true baseline delay of 4.6 or 8.6 days; Erlang-distributed delay with 3 or 10 stages (less variable delays [Lloyd, 2001]); delay defined as the mean time to isolation *among those isolated* ($\tau = 1/D - \gamma$); a population of 10 million; and implementation ramp-up over 30 or 90 days.

2.8 Out-of-sample validation

Using only data to 12 September, we forecast confirmed cases reported by 27 September and 28 October (baseline: nothing changes) and compared them with the official cumulative counts [UK Health Security Agency, 2026, US Centers for Disease Control and Prevention, 2026], together with a benchmark of constant incidence equal to the mean of the last 14 days.

2.9 Software and reproducibility

Everything was run in R 4.3.3 (2024-02-29) with the `deSolve` package (version 1.40) [R Core Team, 2024, Soetaert et al., 2010]; base R provides optimisation, random numbers and graphics. All random seeds are fixed. Eight scripts (`R/00_functions.R` to `R/07_figures_tables.R`) run in order from `R/run_all.R`; `tests/test_model.R` checks the code against analytic results (e.g. $R_e = 1$ exactly when the growth rate is zero; stochastic mean within 5% of the deterministic solution). The scripts write every table, figure and in-text number (as LaTeX macros) used here. The code is listed in Appendix D and the repository README gives run times and the directory layout (Appendix C).

3 Results

3.1 Surveillance data and model fit

Confirmed cases per day rose from about 35–40 in early June to a plateau of about 80 per day from late July (Figure 2). Reports of onset dates reach their final value slowly: the fitted reporting delay has a mean of 8.5 days, so the apparent fall in the last three weeks of the onset curve is a reporting artefact, which the fitted model reproduces (Figure 2 b). A change in transmission was strongly supported in both datasets (Table 3): ΔAIC against constant transmission was 14.5 (onset curve) and 15.4 (report date). Growth was 0.018 (0.014 to 0.022) per day before about 24 July (95% profile interval 18 July to 7 August) and 0.000 (−0.006 to 0.006) per day afterwards in Fit B, corresponding to R_e of 1.20 (1.16 to 1.25) before and 1.00 (0.93 to 1.06) after. Fit A agreed (change about 5 July, interval 28 June to 2 August; growth after 0.004 (−0.001 to 0.006); R_e 1.04 (0.98 to 1.07)). The model's incidence on 12 September was 83 (68 to 100) cases per day (Fit B) or 63 (55 to 71) (Fit A, which covers only cases with a recorded onset date, about 66% of confirmed cases).

Table 3: Fitted parameters (median and 95% uncertainty interval of draws combining the profile over the change date and the Hessian approximation), with isolation effectiveness 80% and baseline delay 7 days. Growth rates are per day. Fits for 50% and 100% give the same growth rates (Appendix Table 11).

Fit	Growth rate before (/d)	Growth rate after (/d)	R_e before	R_e after	Change date (95% interval)	ΔAIC	Incidence 12 Sep (/d)
A: onset curve	0.0169 (0.0112–0.0234)	0.0040 (−0.0015 to 0.0065)	1.19 (1.13–1.27)	1.04 (0.98–1.07)	5 July (28 June–2 August)	14.5	63 (55–71)
B: report date	0.0177 (0.0140–0.0216)	0.0004 (−0.0062 to 0.0058)	1.20 (1.16–1.25)	1.00 (0.93–1.06)	24 July (18 July–7 August)	15.4	83 (68–100)

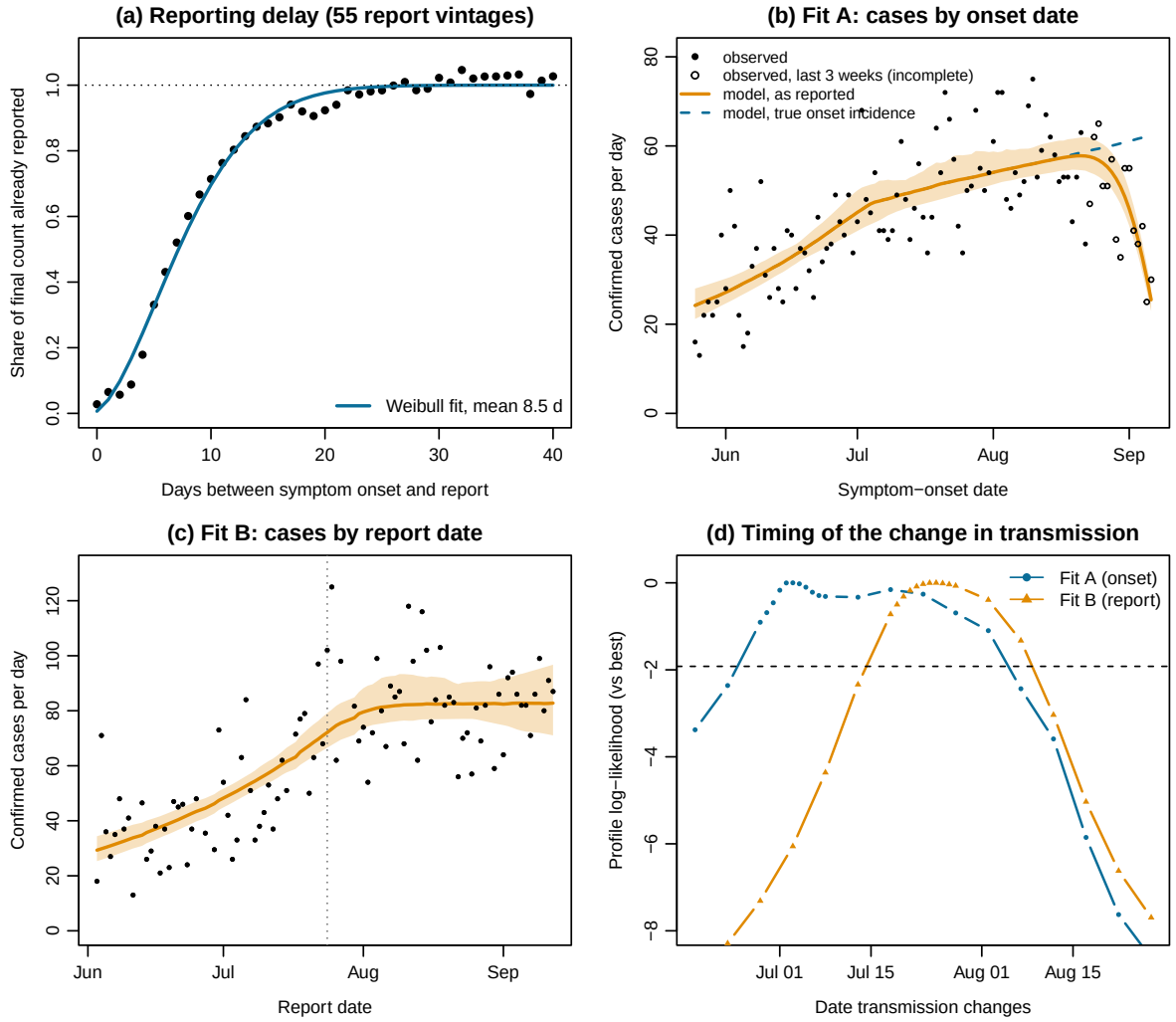


Figure 2: Model fit. (a) Reporting delay measured from 55 SitRep vintages. (b) Fit A: confirmed cases by onset date; open circles are the incomplete last three weeks; orange is the model as it would be reported, blue dashed the true onset incidence it implies. (c) Fit B: confirmed cases per day by report date (22 July step removed); dotted line marks the estimated change date. (d) Profile log-likelihood for the date of the change in transmission (dashed line: 95% limit).

Both fits were adequate: 95% negative-binomial predictive intervals covered 96% (Fit A) and 95% (Fit B) of observations, with lag-1 residual autocorrelation of 0.27 and 0.01. Refitting Fit B at the four extreme corners of the natural-history ranges changed the post-change growth rate only between −0.0005 and 0.0009 per day and the 12 September incidence between 81 and 84 (Appendix Table 10), which supports using these two quantities as data-determined inputs to the scenarios. Over the same period testing expanded from 168 to 494 samples per day and positivity fell from 22.6% to 17.1% (Appendix Table 12): confirmed counts grew much more slowly than testing, so the plateau is unlikely to be purely a laboratory-capacity artefact, although the data cannot show whether true incidence was flat or falling.

3.2 Baseline: a 7-day delay

With the delay unchanged the model projects a plateau: median R_e 1.00, with a proportion 0.44 of draws below 1, and 31,089 (9,845 to 107,831) symptomatic cases in the next 12 months (median and 95% UI). The interval is wide because the uncertainty in the growth rate (about ±0.006 per day) compounds over a year. Transmission had ended within five years in only a proportion 0.12 of draws; in the remainder it persisted.

3.3 Scenarios

Shortening the delay converted the plateau into decline in all eight alternative scenarios (Table 4, Figure 3 and Figure 4). At 80% effectiveness, a 5-day delay averted 73% of cases (≈22,821 cases) and a 3-day delay averted 89% (27,786 cases). The corresponding R_e were 0.89 (0.82 to 0.95) and 0.74 (0.68 to 0.79). At 50% effectiveness the benefit was smaller but still substantial (50% for 5 days; 75% for 3 days), and at 100% it was 84% and 94%. Illustratively, applying the crude fatality ratio of 48%, the 3-day scenario at 80% effectiveness corresponds to about 13,421 fewer deaths.

Table 4: Scenario results from 12 September 2026 (median and 95% UI over 1,000 paired draws). Months to end: first day with fewer than one exposed or infectious person; >60 means not ended within five years in at least half of draws. †Cases averted × 48%; illustrative upper bound.

Isolation effectiveness	Delay (days)	R_e median (95% UI)	Cases in 12 months median (95% UI)	Cases averted (%) median (95% UI)	Deaths averted† median	Months to end median (95% UI)	P(ended ≤24 mo)
50%	7	1.00 (0.92–1.08)	31,201 (9,854–109,025)	–	–	>60	0.01
	5	0.95 (0.86–1.02)	15,405 (6,232–44,057)	50 (33–62)	7,532	53 (21 to >60)	0.05
	3	0.87 (0.80–0.93)	7,695 (4,033–16,443)	75 (57–86)	11,346	22 (13 to 49)	0.60
80%	7	1.00 (0.92–1.07)	31,089 (9,845–107,831)	–	–	>60	0.01
	5	0.89 (0.82–0.95)	8,212 (4,127–18,244)	73 (55–84)	11,023	24 (14 to 53)	0.52
	3	0.74 (0.68–0.79)	3,313 (2,133–5,309)	89 (76–95)	13,421	9 (7 to 14)	1.00
100%	7	1.00 (0.93–1.06)	30,961 (9,833–106,530)	–	–	>60	0.01
	5	0.83 (0.77–0.88)	4,807 (2,779–8,746)	84 (68–93)	12,588	13 (9 to 21)	0.99
	3	0.60 (0.55–0.64)	1,817 (1,234–2,728)	94 (85–98)	14,050	5 (4 to 7)	1.00

Time to the end of transmission was even more sensitive to the delay than cases. At 80% effectiveness the median was 9 months (UI 7–14) at 3 days against 24 months (14–53) at 5 days and more than 60 months at 7 days (Figure 4 b). With only 50% effectiveness a 3-day delay still ended transmission in a median of 22 months, whereas a 5-day delay needed 53 months (UI 21 to >60). The probability that transmission had ended within two years was 1.00 (3 days) and 0.52 (5 days) at 80% effectiveness, and 0.60 and 0.05 at 50%.

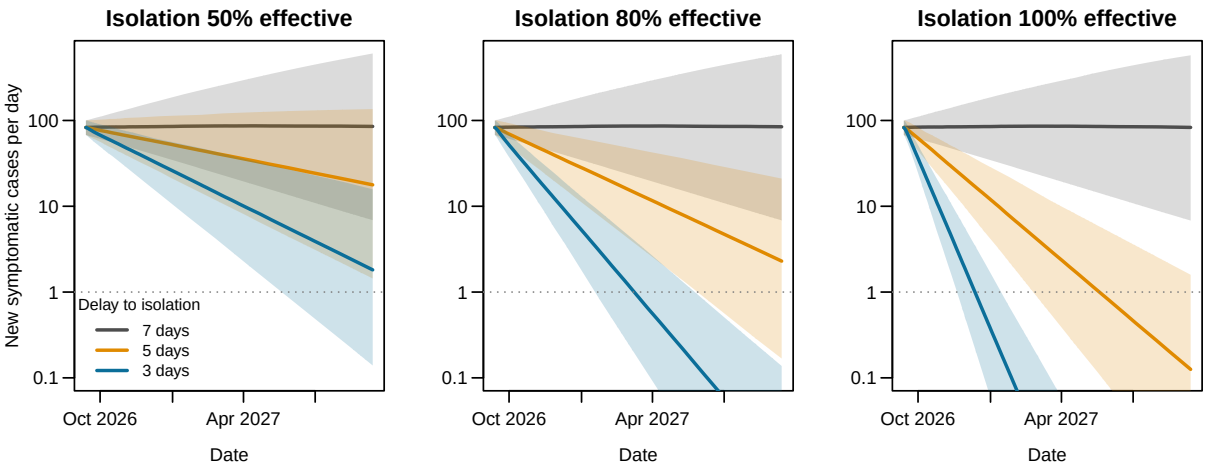


Figure 3: Projected new symptomatic cases per day from 12 September 2026 for the three delays (median and 95% UI; logarithmic scale; dotted line: one case per day) at each isolation effectiveness.

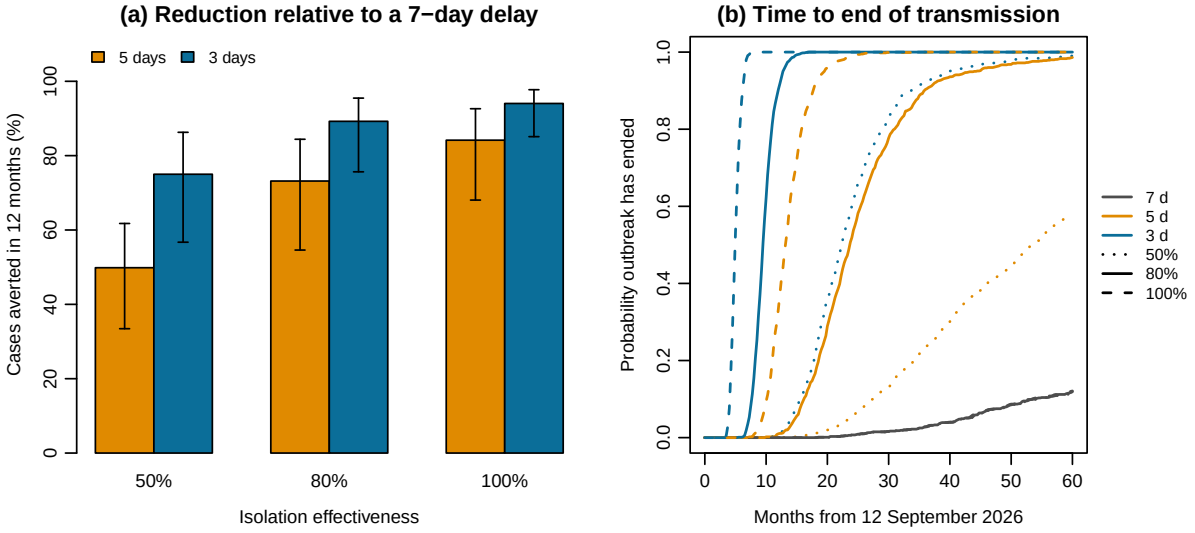


Figure 4: (a) Percentage of 12-month cases averted relative to the 7-day delay with the same isolation effectiveness (median, 95% UI). (b) Probability that transmission has ended, by months from 12 September 2026, for the nine scenarios (colour: delay; line type: effectiveness).

The continuous surface (Figure 5) shows diminishing returns: at 80% effectiveness the first two days gained (7 to 5) averted 73% and the next two (5 to 3) a further 16 percentage points, and delays above 7 days increase cases steeply. Because the baseline sits at the threshold, the white contour (half of draws with $R_e < 1$) lies at the 7-day delay itself.

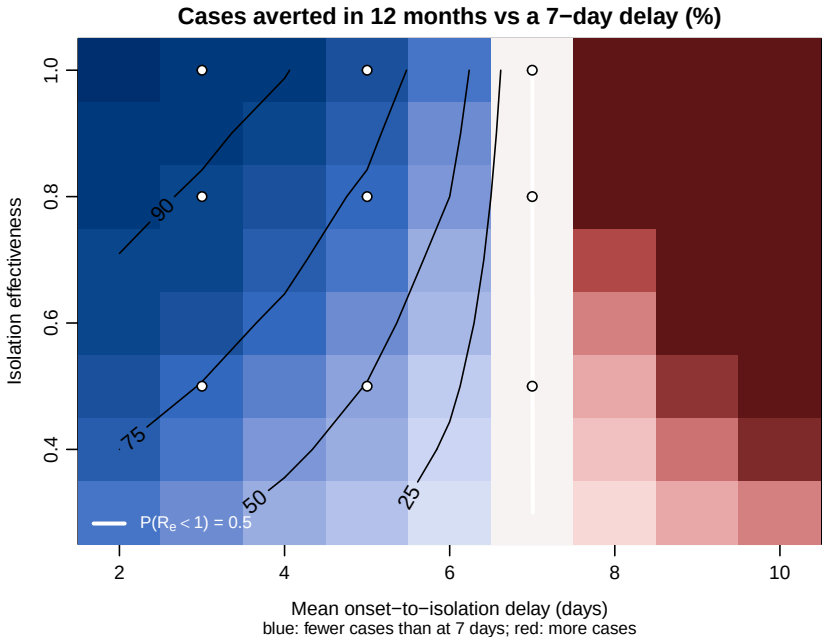


Figure 5: Policy surface: median percentage of 12-month cases averted relative to a 7-day delay, by delay and isolation effectiveness (100 draws per cell). Circles mark the nine scenarios; the white line marks where half of draws have $R_e < 1$.

3.4 Stochastic check

The stochastic model reproduced the deterministic medians (ratio 1.00 to 1.09 across scenarios; Table 5, Figure 6). At 80% effectiveness, median cases fell from 32,114 (7 days) to 8,904 (5 days) and 3,588 (3 days), reductions of 72% and 89%. Chance events made early extinction more likely than in the deterministic model: the probability of ending within five years at 7 days was 0.31 (deterministic 0.12), and within two years at 5 days 0.73 (deterministic 0.52). The conclusions are therefore robust to demographic stochasticity.

Table 5: Stochastic (500 replicates) and deterministic results by scenario.

Isolation effectiveness	Delay (days)	Stochastic cases median (95% range)	Deterministic median	P(ended ≤12 mo)	P(ended ≤24 mo)	P(ended ≤60 mo)	Deterministic P(≤60 mo)
50%	7	31,982 (8,450–107,440)	31,554	0.00	0.14	0.27	0.12
	5	16,508 (5,864–46,870)	15,776	0.00	0.27	0.71	0.56
	3	8,528 (3,956–19,595)	7,822	0.08	0.71	0.99	0.99
80%	7	32,114 (8,623–110,110)	31,391	0.00	0.14	0.31	0.12
	5	8,904 (4,071–19,945)	8,281	0.10	0.73	0.99	0.98
	3	3,588 (2,173–5,772)	3,313	0.86	1.00	1.00	1.00
100%	7	31,062 (8,615–107,137)	31,147	0.01	0.16	0.36	0.12
	5	5,055 (2,731–9,547)	4,792	0.61	1.00	1.00	1.00
	3	1,914 (1,279–2,893)	1,815	1.00	1.00	1.00	1.00

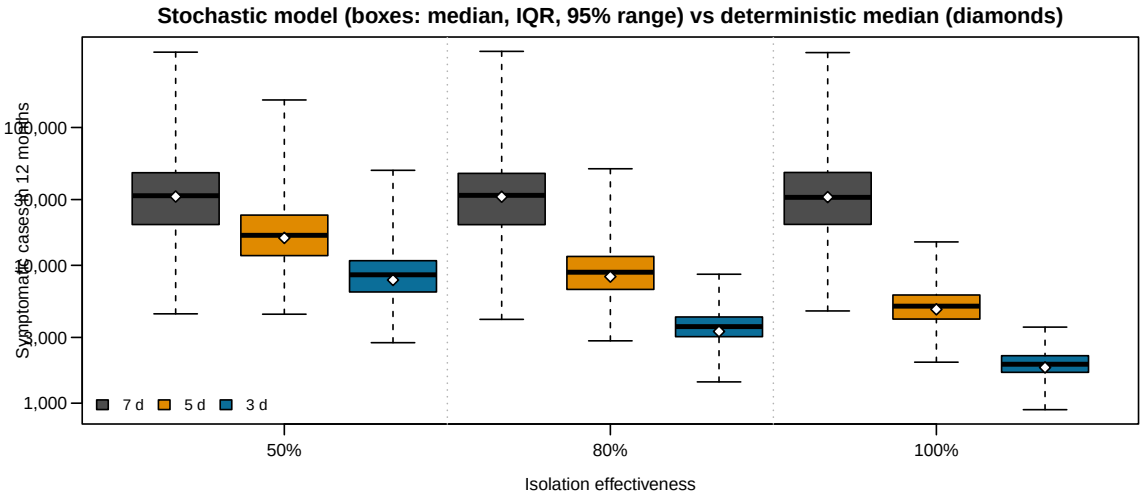


Figure 6: Stochastic distribution of 12-month cases (boxes: median and interquartile range; whiskers: 95% range) with the deterministic median (diamonds).

3.5 Sensitivity analyses

In the global analysis (Figure 7 a; Appendix Table 9) the isolation delay had the largest influence on 12-month cases (PRCC 0.94), followed by isolation effectiveness (−0.80), baseline growth rate (0.42), latent period (0.27), starting incidence (0.17) and infectious period (0.10). In the structural and assumption analyses (Figure 7 b; Table 6), percentage reductions at 80% effectiveness ranged from 56 to 84% for a 5-day delay and 77 to 96% for a 3-day delay, with one exception. Percentages were almost unchanged across the 95% range of starting incidence (73 to 73% for 5 days), although absolute counts scaled with it. Reductions were larger with less variable delays (Erlang-3: 77%; Erlang-10: 78% at 5 days), and also when the delay was defined among isolated cases (83% and 93%), because in that reading the 7-day baseline isolates only 6% of cases. They were smaller when interventions took 30 or 90 days to reach full effect (69% and 62% for 5 days; 86% and 81% for 3 days), and were insensitive to population size (73%). Fitting to the onset curve gave 82% and 94% but a larger baseline case count (51,067) because its post-change growth was slightly positive. The baseline growth rate was the main source of spread in the baseline projection, ranging from 11,750 to 92,275 cases across its 95% range, while the percentage averted moved from 56 to 84% (5 days).

The exception is the baseline delay. If the true current mean delay were already 4.6 days (the preprint's onset-to-sample mean), a "5-day" policy would be *worse* than the status quo: cases would rise (percentage averted: −67 (−98 to −37)), while a 3-day target would still avert 80 (63 to 90) percent. If the current delay were 8.6 days, the gains would be larger (5 days: 82%; 3 days: 91%). Hence the 7-day baseline must be verified.

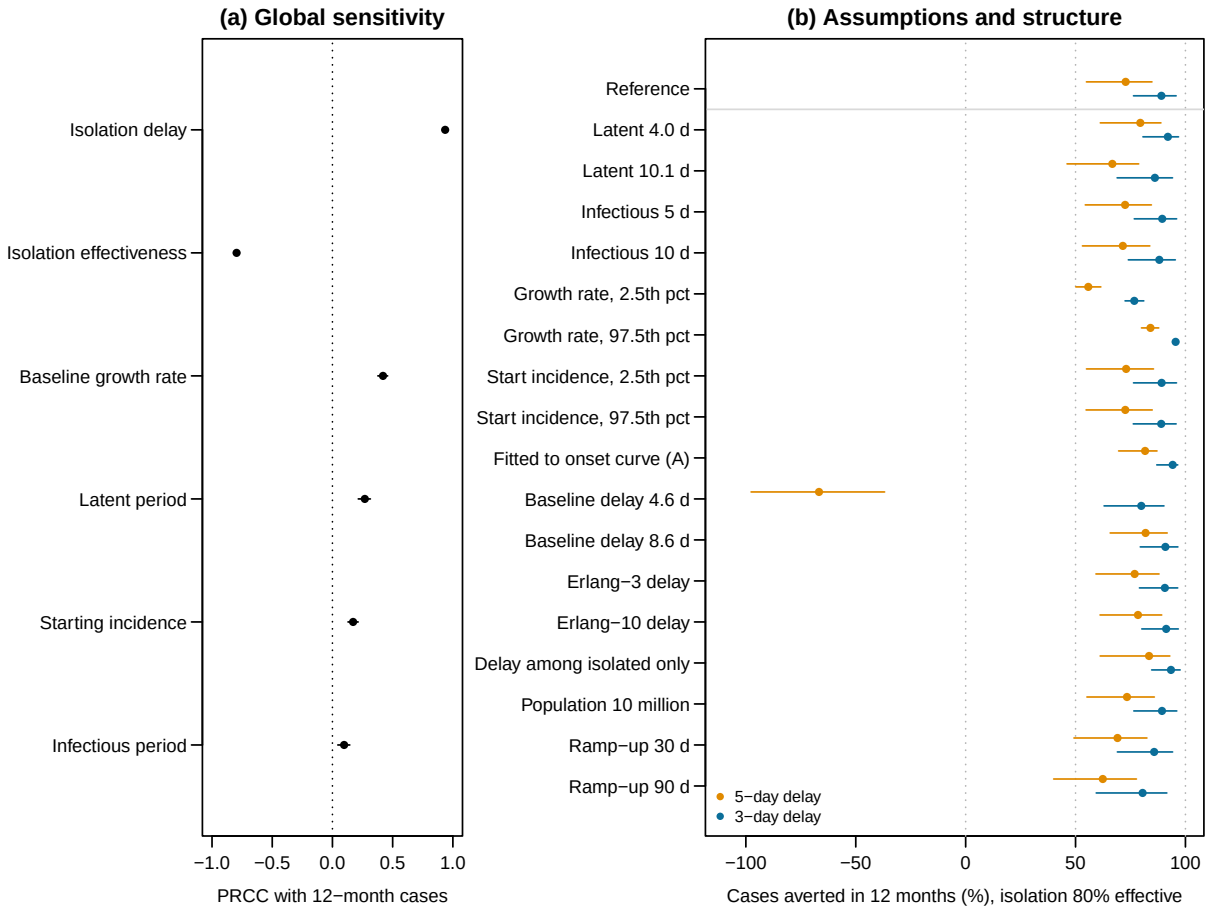


Figure 7: (a) PRCC (with 95% bootstrap CI) of 12-month cases against each parameter. (b) Percentage of 12-month cases averted at 80% effectiveness under each assumption (median and 95% UI), for 5-day and 3-day delays.

Table 6: Assumption and structure analyses at 80% isolation effectiveness: percentage of 12-month cases averted vs the (variant-specific) baseline delay; median (95% UI) over 300 draws.

Variant	5-day delay (%)	3-day delay (%)	R_e at 3 d
Reference (all uncertainty propagated)	73 (55 to 85)	89 (76 to 96)	0.74
Natural history			
Latent period 4.0 d	79 (61 to 89)	92 (81 to 97)	0.74
Latent period 10.1 d	67 (46 to 79)	86 (69 to 94)	0.74
Infectious period 5 d	73 (54 to 85)	89 (77 to 96)	0.75
Infectious period 10 d	72 (53 to 84)	88 (74 to 95)	0.73
Calibration			
Growth rate at 2.5th percentile	56 (50 to 62)	77 (73 to 81)	0.68
Growth rate at 97.5th percentile	84 (80 to 88)	96 (94 to 97)	0.78
Starting incidence at 2.5th percentile	73 (55 to 85)	89 (76 to 96)	0.74
Starting incidence at 97.5th percentile	73 (55 to 85)	89 (76 to 96)	0.74
Calibrated to onset curve (Fit A) instead of report counts	82 (70 to 87)	94 (87 to 96)	0.77
Baseline delay			
True baseline delay 4.6 d (onset-to-sample, preprint)	−67 (−98 to −37)	80 (63 to 90)	0.85
True baseline delay 8.6 d (onset-to-result, preprint)	82 (66 to 92)	91 (79 to 97)	0.69
Structure			
Erlang-3 delay (less variable)	77 (59 to 88)	91 (79 to 96)	0.70
Erlang-10 delay (nearly fixed delay)	78 (61 to 89)	91 (80 to 97)	0.69
Delay = mean time to isolation among those isolated	83 (61 to 93)	93 (85 to 98)	0.56
Population at risk 10 million	73 (55 to 86)	89 (77 to 96)	0.74
Implementation			
Interventions ramp up linearly over 30 d	69 (49 to 82)	86 (69 to 94)	0.74
Interventions ramp up linearly over 90 d	62 (40 to 78)	81 (59 to 92)	0.74

3.6 Out-of-sample validation

Forecasts made with data to 12 September overestimated subsequent cases (Table 7, Figure 8). Over the next 15 days the model projected 1,236 new confirmed cases against 916 observed (error 35%; observed outside the model's 95% UI), and over 46 days 3,781 (2,796 to 4,991) (median and 95% UI) against 3,074 observed (error 23%; observed inside the UI). Observed incidence fell to about 61 per day by 27 September and averaged 67 per day over 46 days, against about 83 per day on 12 September. The constant-rate benchmark erred in the same direction (24% at 46 days). The model is therefore no worse than the simplest alternative, but the later data indicate a modest decline after 12 September.

Table 7: Out-of-sample forecast of new confirmed cases since 12 September (data to that date only). UI: parameter uncertainty; predictive interval adds negative-binomial noise. Error: (forecast − observed)/observed.

Horizon	Observed new	Model median	Model 95% UI	Predictive 95% interval	Model error	Constant-rate error
15 d (27 September)	916	1,236	1,017–1,490	993–1,511	35%	36%
46 d (28 October)	3,074	3,781	2,796–4,991	2,743–5,126	23%	24%

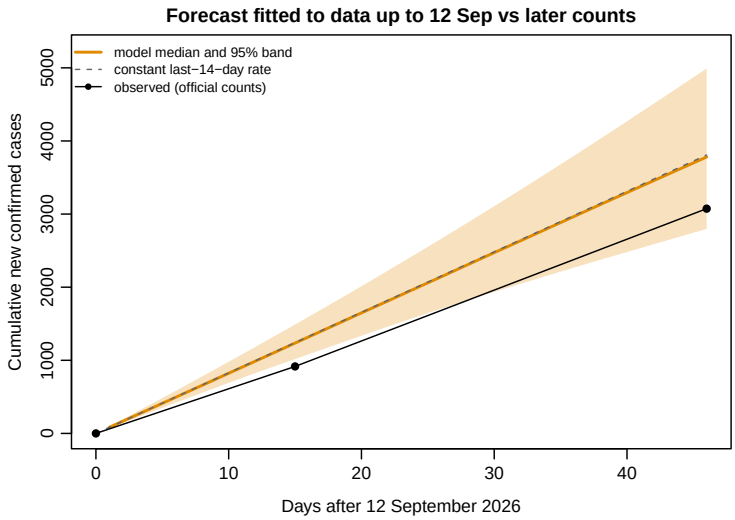


Figure 8: Forecast of cumulative new confirmed cases after 12 September 2026 against the official counts on 27 September and 28 October.

4 Discussion

Principal findings. In a model anchored to the observed course of the DRC outbreak, shortening the mean onset-to-isolation delay from 7 to 5 or 3 days turns a plateau into a decline, avoiding most of the next year's cases in every scenario tested, provided that the current delay really is about 7 days. At 80% isolation effectiveness the reductions were 73% and 89%, R_e fell to 0.89 and 0.74, and the median end of transmission moved from beyond five years to about 24 and 9 months. Speed and quality both matter: even a 3-day delay left 22 months of transmission if isolation was only 50% effective, and PRCC for effectiveness was nearly as large as for the delay.

Why the effect is large, and when it would not be. The effect on cumulative cases is large because the outbreak is close to threshold. A shorter delay raises the share of cases isolated before removal (from 50% to 58% and 70%), which lowers R_e by a fixed proportion: from Eq. (2), R_e is multiplied by $(1 - \epsilon a_{\text{new}})/(1 - \epsilon a_7)$ with $a = \tau/(\tau + \gamma)$. At $\epsilon = 0.8$ this factor is 0.89 for a 5-day delay and 0.73 for a 3-day delay. When baseline $R_e \approx 1$ any such factor produces exponential decline; when baseline R_e is higher it does not. Table 8 gives the largest baseline R_e that each delay could bring below 1. Before the July slowdown R_e was about 1.20; a 3-day delay with 80% or better isolation would then have been sufficient (threshold 1.36), a 5-day delay would not (1.13). Percentage reductions over 12 months should therefore not be transferred to settings or periods with a higher R_e.

Table 8: Largest baseline reproduction number (with a 7-day delay) that a new delay can bring below 1, from Eq. (2) with $\gamma = 1/7$.

Isolation effectiveness	5-day delay	3-day delay
50%	1.06	1.15
80%	1.13	1.36
100%	1.20	1.67

What "delay" means. In an exponential model with competing removal, the mean delay is not the mean time to isolation of those who are isolated: with a 7-day delay and a 7-day infectious period only about half of cases are isolated at all. A field-measured onset-to-isolation or onset-to-sample delay is conditional on being isolated, so the two are not interchangeable. The structural analyses bracket this: less variable delays and the conditional definition give larger benefits (up to 83% and 93%), not smaller, so the base-case figures are not extreme in either direction on this point. A related point is that infectiousness of Ebola disease changes with time since onset, which the model ignores.

Operational implications. The results support effort aimed at shortening the delay, with the caveat that benefits depend on isolation actually preventing contact: the most effective route would combine faster detection (community alert networks, decentralised sampling, rapid result return, which the preprint puts at 8.6 days from onset to test result [Ebengo et al., 2026]) with isolation capacity and quality (occupancy was above 100% in some facilities and 59% of confirmed deaths occurred outside treatment units [Kabasele et al., 2026]). Gains diminish: the second two days of improvement add less than the first two. The implementation ramp-up analysis shows that a 90-day roll-out would reduce the 5-day benefit from 73% to 62% (lower limit 40%).

Comparison with the literature. Our qualitative conclusion that both speed and effectiveness of isolation determine control is consistent with the Ebola review of Chowell and Nishiura [Chowell and Nishiura, 2014], with earlier analyses of controllability [Fraser et al., 2004, Lipsitch et al., 2003], and with West African analyses linking faster isolation and community care to reduced transmission [Pandey et al., 2014, Kucharski et al., 2015, Faye et al., 2015]. Projections for the present outbreak by Mooring et al. varied the *fraction* isolated with a 2-day delay and perfect isolation [Mooring et al., 2026]; ours instead varies the delay itself with imperfect isolation and is anchored to the observed surveillance. Our baseline reproduction numbers (1.20 rising phase; 1.00 plateau) are lower than the 2.5–3.5 estimates reported early in the outbreak by branching-process and spatial models [Mooring et al., 2026, Ebengo et al., 2026], consistent with substantial control having already occurred; they are within the 1.3–2.7 range of earlier Central African outbreaks [Chowell and Nishiura, 2014, Lekone and Finkenstädt, 2006, Legrand et al., 2007].

Strengths. The analysis is anchored to the outbreak's own data, accounts for right-truncation and a retrospective data reconciliation, propagates fit and literature uncertainty jointly in paired scenarios, checks conclusions with a stochastic model, global and structural sensitivity analyses

and an out-of-sample forecast, and is fully reproducible from raw data with unit-tested code.

Limitations. (1) *Delay structure.* The exponential competing-risk delay and constant infectiousness are simplifications; the structural analyses suggest conclusions are conservative in the direction tested, but infectiousness rising with time since onset, and post-mortem transmission at unsafe burials, are not modelled. (2) *Baseline delay.* The 7-day baseline is taken from the study brief and could not be confirmed in the primary source; shorter true delays would reverse the benefit of a 5-day target. (3) *Identifiability.* Case counts constrain growth, not ϵ, the delay or the natural history; ϵ is a scenario assumption and the literature ranges are not drawn from the Nash database. (4) *Data.* Confirmed cases depend on testing, definitions and harmonisation; the onset curve is digitised and covers about two thirds of cases; we treated national data as homogeneous, whereas provinces differ (declining in Ituri, rising in North Kivu [UK Health Security Agency, 2026]). (5) *Forecast error.* The out-of-sample forecast overestimated by 23% at 46 days; absolute counts are conditional on the plateau at 12 September, although percentage effects were insensitive to starting incidence. (6) *Omitted processes.* No contact tracing, health-worker infections, spatial structure, isolation-capacity constraints, behaviour change, or changes in fatality ratio; deaths averted are illustrative. (7) *Horizon and threshold.* Percentages depend on the 12-month horizon and on baseline $R_e \approx 1$. (8) *Counterfactual timing.* Scenarios begin on 12 September; the actual course since then is only used for validation.

Further work. (i) Measure onset-to-isolation delays directly (line list); (ii) model the delay as a gamma-distributed waiting time with explicit isolated fraction; (iii) add contact tracing, health-care and funeral transmission; (iv) fit province-specific transmission; (v) refit with updated data to 28 October and beyond; (vi) replace literature ranges by the Nash et al. parameter database; (vii) add costs of delay-reduction options.

5 Conclusion

For a model anchored to the observed course of the 2026 BVD outbreak, reducing the mean onset-to-isolation delay from 7 to 5 or 3 days would avert most cases over the next year and bring forward the end of transmission by years, provided isolation is effective and the true current delay is about 7 days. The principal uncertainties are the true current delay, the effectiveness of isolation and the baseline growth rate, all of which can be reduced by measurement. We recommend that onset-to-isolation delay be measured and reported routinely.

Declarations

Author contributions. All authors contributed equally to the project (conception, modelling, analysis and writing); individual CRediT roles to be specified by the authors. **Funding.** None declared (to be confirmed). **Competing interests.** None declared (to be confirmed). **Ethics.** Only aggregated, publicly available surveillance data were used. **Data and code availability.** Surveillance data are in `data/bvd_2026/` (public sources, see their README); the analysis code,

tests, results and this manuscript's source are in the accompanying repository; `R/run_all.R` regenerates every number, table and figure. **Use of generative AI.** Generative AI (Claude, Anthropic) was used for brainstorming, structuring the modelling workflow, explaining technical concepts, writing and testing analysis code, and drafting and language editing of this manuscript. The authors are responsible for reviewing and approving all analyses, interpretations and written content, and for verifying the references. **Acknowledgements.** We thank our advisers Dr James Azam and Andrej Jarynowski, and the INSP, INRB-UMIE and BVDOutbreakSize teams whose public data made this analysis possible.

References

Delayo L. Zomahoun, Mary Adetinuke Boyd, Margaret A. Honein, Joel M. Montgomery, Emily Zielinski-Gutierrez, Elissa Meites, Athalia Christie, and Satish K. Pillai. Outbreak of Ebola disease caused by Bundibugyo virus — Democratic Republic of the Congo and Uganda, may 2026. *MMWR Morb Mortal Wkly Rep*, 75:293–294, 2026. doi: 10.15585/mmwr.mm7522e3.

Dumazedier Kabasele, Erika Meyer, Issaka Kabore, Amber Dismer, et al. Characteristics and monitoring of the 2026 outbreak of Ebola disease caused by Bundibugyo virus — Democratic Republic of the Congo, August 2026. *MMWR Morb Mortal Wkly Rep*, 75(35):554–556, 2026. doi: 10.15585/mmwr.mm7535e1.

US Centers for Disease Control and Prevention. Ebola outbreak: current situation. https://www.cdc.gov/ebola/situation-summary/index.html, 2026. Data as of 28 October 2026; accessed 31 October 2026.

Dav M. Ebengo, Pierre Akilimali, et al. Early epidemiological and spatial assessment of the Bundibugyo Ebola virus outbreak 2026. Epidemiological forum (preprint, not peer reviewed), 2026. Posted 12 June 2026.

Christophe Fraser, Steven Riley, Roy M. Anderson, and Neil M. Ferguson. Factors that make an infectious disease outbreak controllable. *Proceedings of the National Academy of Sciences*, 101 (16):6146–6151, 2004.

Gerardo Chowell and Hiroshi Nishiura. Transmission dynamics and control of Ebola virus disease (EVD): a review. *BMC Medicine*, 12:196, 2014. doi: 10.1186/s12916-014-0196-0.

Abhishek Pandey, Katherine E. Atkins, Jan Medlock, et al. Strategies for containing Ebola in West Africa. *Science*, 346:991–995, 2014.

Adam J. Kucharski, Anton Camacho, Francesco Checchi, et al. Measuring the impact of Ebola control measures in Sierra Leone. *Proceedings of the National Academy of Sciences*, 112(46): 14366–14371, 2015.

Eric Q. Mooring, William T. Koval, Isobel Routledge, Inga Holmdahl, Kristen Hudson, Guido España, Rebecca Kahn, and Beau B. Bruce. Modeled scenario projections for the Ebola disease

outbreak caused by Bundibugyo virus, 2026. *MMWR Morb Mortal Wkly Rep*, 75(22):285–289, 2026. doi: 10.15585/mmwr.mm7522e1.

Sam Abbott, Katharine Sherratt, et al. BVDOutbreakSize: analysis and cleaned surveillance streams for the 2026 Bundibugyo virus disease outbreak. https://github.com/epiforecasts/BVDOutbreakSize, 2026. Data as of 12 September 2026; underlying sources: INSP situation reports and INRB-UMIE transcriptions.

UK Health Security Agency. Outbreaks under monitoring in 2026, weeks 36 and 39. https://www.gov.uk/government/publications/outbreaks-under-monitoring-in-2026, 2026. Accessed 31 October 2026.

Rebecca K. Nash et al. Ebola virus disease mathematical models and epidemiological parameters: a systematic review. *Lancet Infectious Diseases*, 2024. PMID 39127058; preprint doi 10.1101/2024.03.20.24304571.

S. M. Blower and H. Dowlatabadi. Sensitivity and uncertainty analysis of complex models of disease transmission: an HIV model, as an example. *International Statistical Review*, 62(2): 229–243, 1994.

Simeone Marino, Ian B. Hogue, Christian J. Ray, and Denise E. Kirschner. A methodology for performing global uncertainty and sensitivity analysis in systems biology. *Journal of Theoretical Biology*, 254(1):178–196, 2008.

Alun L. Lloyd. Realistic distributions of infectious periods in epidemic models: changing patterns of persistence and dynamics. *Theoretical Population Biology*, 60(1):59–71, 2001.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2024. URL https://www.R-project.org/.

Karline Soetaert, Thomas Petzoldt, and R. Woodrow Setzer. Solving differential equations in R: package deSolve. *Journal of Statistical Software*, 33(9):1–25, 2010.

Marc Lipsitch, Ted Cohen, Ben Cooper, et al. Transmission dynamics and control of severe acute respiratory syndrome. *Science*, 300:1966–1970, 2003.

Ousmane Faye et al. Chains of transmission and control of Ebola virus disease in Conakry, Guinea, in 2014: an observational study. *Lancet Infectious Diseases*, 15(3):320–326, 2015.

Phenyo E. Lekone and Bärbel F. Finkenstädt. Statistical inference in a stochastic epidemic SEIR model with control intervention: Ebola as a case study. *Biometrics*, 62(4):1170–1177, 2006.

J. Legrand, R. F. Grais, P. Y. Boelle, A. J. Valleron, and A. Flahault. Understanding the dynamics of Ebola epidemics. *Epidemiology and Infection*, 135(4):610–621, 2007.

P. van den Driessche and James Watmough. Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission. *Mathematical Biosciences*, 180: 29–48, 2002.

J. Wallinga and M. Lipsitch. How generation intervals shape the relationship between growth rates and reproductive numbers. *Proceedings of the Royal Society B*, 274:599–604, 2007.

A Mathematical details

Reproduction number. For Model (1) with $S = N$, one case in I spends $1/(\tau+\gamma)$ days non-isolated and is isolated with probability $a = \tau/(\tau+\gamma)$, after which it spends a further $1/\gamma$ days at relative infectiousness $1-\epsilon$. Hence $R_e = \beta\{1/(\tau+\gamma) + a(1-\epsilon)/\gamma\} = (\beta/\gamma)(1-\epsilon a)$, which is Eq. (2) (next-generation method [van den Driessche and Watmough, 2002]).

Erlang delay. Replacing I by m stages $I_1, \ldots, I_m$ with progression rate $m\tau$ and removal γ from each, and isolation on leaving I_m, gives a^m with $a = m\tau/(m\tau+\gamma)$ as the isolated share and $R_e = (\beta/\gamma)(1-\epsilon a^m)$; as $m \to \infty$ the delay is fixed and the isolated share is $e^{-\gamma D}$ (0.37 for $D = \gamma^{-1} = 7$). This is used for the Erlang-3 and Erlang-10 analyses (function `make_rhs`).

Calibration. For fixed $(\sigma, \gamma, \tau_0, \epsilon)$ the growth rate of the linearised system with state $(E, I_1..I_m, Q)$ is the dominant eigenvalue of its Jacobian; β is the root of $\mathrm{eig}(\beta) = r$ (`uniroot`). The relation between growth rate and R_e depends on the generation-time distribution [Wallinga and Lipsitch, 2007], which is why β is recalibrated per draw. In the stochastic model the same is done with the one-step expected-value matrix.

Initial state. The scenario starts from the dominant eigenvector (the stable distribution across E, I, Q at the baseline), scaled so that $\sigma E = i(T)$ (the fitted incidence on 12 September).

Reporting delay. With $c(L) = 1-\exp\{-((L+0.5)/s)^\kappa\}$ the daily delay probability mass is $p(L) = c(L) - c(L-1)$, $c(-1) = 0$. Fit B uses expected cumulative reports $\hat{C}(u) = \sum_{v \le u} \sum_{d \le v} i(d)p(v-d)$ and increments $\hat{C}(u_j) - \hat{C}(u_{j-1})$ over report intervals.

Stochastic model. Per step Δt: $n_{SE} \sim \mathrm{Bin}(S, 1-e^{-\lambda\Delta t})$, $n_{EI} \sim \mathrm{Bin}(E, 1-e^{-\sigma\Delta t})$, $n_{IQ} \sim \mathrm{Bin}(I, 1-e^{-\tau\Delta t})$, then removals $\mathrm{Bin}(I-n_{IQ}, 1-e^{-\gamma\Delta t})$ and $\mathrm{Bin}(Q, 1-e^{-\gamma\Delta t})$.

PRCC. Ranks of each input and of the output are regressed on the ranks of the other inputs; the PRCC is the correlation of the residuals [Blower and Dowlatabadi, 1994, Marino et al., 2008].

B Supplementary tables

Table 9: Partial rank correlation of 12-month cases with each parameter.

Parameter (range sampled)	PRCC (95% bootstrap CI)
Isolation delay	0.94 (0.93 to 0.94)
Isolation effectiveness	-0.80 (-0.82 to -0.77)
Baseline growth rate	0.42 (0.38 to 0.46)
Latent period	0.27 (0.21 to 0.32)
Starting incidence	0.17 (0.13 to 0.21)
Infectious period	0.10 (0.04 to 0.14)

Table 10: Fit B refitted at extreme natural-history values (change date fixed at the best estimate; $\epsilon = 0.8$).

Latent (d)	Infectious (d)	Neg. log-lik.	β after	Growth rate after (/d)	Incidence 12 Sep (/d)
4.0	5	407.60	0.300	-0.0001	81.7
10.1	5	407.59	0.298	-0.0005	81.2
4.0	10	407.62	0.191	0.0009	83.5
10.1	10	407.58	0.190	0.0002	82.4

Table 11: Transmission rate and reproduction number by isolation effectiveness (refit at the best change date). The growth rates, which are what the data identify, do not depend on ϵ.

Fit	Isolation eff.	β before	β after	Growth after (/d)	R_e before	R_e after
A	50%	0.235	0.201	0.0046	1.24	1.06
B	50%	0.233	0.191	0.0003	1.23	1.00
A	80%	0.288	0.250	0.0044	1.21	1.05
B	80%	0.286	0.238	0.0000	1.20	1.00
A	100%	0.339	0.298	0.0043	1.19	1.04
B	100%	0.337	0.285	-0.0003	1.18	1.00

Table 12: Testing and positivity by month, 2026 (new confirmed cases after removing the 22 July step; samples analysed per day interpolated over missing days).

Month (2026)	New confirmed cases	Samples analysed	Samples per day	Positivity (%)
Jun	987	4,365	168	22.6
Jul	1,996	9,102	294	21.9
Aug	2,512	13,373	431	18.8
Sep	1,014	5,927	494	17.1

C Repository layout and how to reproduce

```
taus-isolation-delay/
  README.md              how to run; run times; caveats
  data/
    bvd_2026/            surveillance streams (public; its README states sources and caveats
    cumulative_confirmed_drc.csv   later official counts used only for validation
  R/
    00_functions.R       model, calibration, stochastic engine, helpers, constants (CFG)
    01_data_prep.R       reporting delay and the two fitting datasets
    02_fit.R             maximum likelihood fit, profile over change date, draws, adequacy
    03_scenarios.R       nine scenarios x paired draws; policy surface
    04_stochastic.R      stochastic check
```

```
  05_sensitivity.R      PRCC and structural variants
  06_validation.R       out-of-sample forecast
  07_figures_tables.R   figures, LaTeX tables, number macros (manuscript/numbers.tex)
  run_all.R             runs tests, scripts 01-07, then compiles the manuscript
tests/test_model.R      unit tests of the model code
manuscript/             main.tex, references.bib, numbers.tex (auto-generated)
results/, figures/      all outputs (regenerated)
```

To reproduce: install R (here 4.3.3 (2024-02-29)) and the `deSolve` package, then run `Rscript R/run_all.R` from the repository root (about 15 minutes) and, to rebuild the PDF, `pdflatex`/`bibtex` in `manuscript/`. Random seeds are set at the top of each script. The test script must pass before any analysis is run.

D R code

The scripts below are inserted directly from the repository files so the listing always matches the analysis. (`R/07_figures_tables.R`, which only draws figures and writes tables and macros, and `R/run_all.R` are in the repository.)

R/00_functions.R

```r
## 00_functions.R -- shared model code, helpers and constants (sourced by every script)
## Model: S-E-I-Q-R with an isolated compartment Q. The infectious (not yet isolated) class is split into
## m Erlang stages (m = 1 gives the plain exponential delay used in the main analysis; m > 1 is a
## structural sensitivity analysis). Isolation occurs on leaving stage m at rate m*tau (tau = 1/mean delay).
suppressPackageStartupMessages(library(deSolve))

## ------------------------------------------------------------------ paths
find_root <- function() {
  a <- commandArgs(FALSE); f <- sub("--file=", "", a[grep("--file=", a)])
  if (length(f)) normalizePath(file.path(dirname(f[1]), "..")) else normalizePath(".")
}
ROOT <- find_root()
P <- list(data = file.path(ROOT, "data"), bvd = file.path(ROOT, "data", "bvd_2026"),
          res = file.path(ROOT, "results"), fig = file.path(ROOT, "figures"),
          tab = file.path(ROOT, "results", "tables"))
for (d in c(P$res, P$fig, P$tab)) dir.create(d, FALSE, TRUE)

## ------------------------------------------------------------------ constants (single source of truth)
CFG <- list(
  sigma0 = 1 / 6.3,       # central latent period 6.3 d           (Ebengo, Akilimali et al. 2026 model input)
  gamma0 = 1 / 7,         # central infectious period 7 d         (idem; 6.5 d in Chowell & Nishiura 2014)
  lat_tri = c(4.0, 6.3, 10.1),   # triangular (min, mode, max) latent period, days
  inf_tri = c(5.0, 7.0, 10.0),   # triangular (min, mode, max) infectious period, days
  tau0 = 1 / 7,           # baseline onset-to-isolation delay 7 d (study brief)
  delays = c(7, 5, 3),    # scenario delays, days
  eps = c(0.5, 0.8, 1.0), # isolation effectiveness: worst / base / best
  eps_base = 0.8,
  pop = 4392200,          # Ituri population (scalars file); depletion negligible
  t0 = as.Date("2026-05-02"),      # model start for the fit
  cutoff = as.Date("2026-09-12"),  # last day of the surveillance data = start of scenarios
  horizon = 365,          # 12-month horizon for cumulative cases (days)
  max_days = 1825,        # 5-year window for outbreak duration
  cfr = 0.483             # 3,475 / 7,200 confirmed deaths / cases at 12 Sep (upper bound; README caveat 7)
)

## ------------------------------------------------------------------ small helpers
rtri <- function(n, p) {                    # triangular(min, mode, max) by inverse CDF
  u <- runif(n); a <- p[1]; c <- p[2]; b <- p[3]; Fc <- (c - a) / (b - a)
```

```r
  ifelse(u < Fc, a + sqrt(u * (b - a) * (c - a)), b - sqrt((1 - u) * (b - a) * (b - c)))
}
q3 <- function(x, na.rm = TRUE) quantile(x, c(.5, .025, .975), na.rm = na.rm, names = FALSE, type = 1)
fmt <- function(x, d = 0, big = TRUE) formatC(x, format = "f", digits = d, big.mark = if (big) "," else "")
mci <- function(x, d = 0, big = TRUE) { q <- q3(x); sprintf("%s (%s--%s)", fmt(q[1], d, big), fmt(q[2], d, big), fmt(q[3], d, big)) }
tex_escape_pct <- function(s) gsub("%", "\\\\%", s)

## ------------------------------------------------------------------ deterministic model
## NB: layout of the state vector for the Jacobian is (E, I_1..I_m, Q); S, R and the case counter are not needed.
jac_m <- function(beta, sigma, gamma, tau, eps, m = 1) {
  n <- m + 2; J <- matrix(0, n, n); r <- m * tau
  J[1, 1] <- -sigma; J[1, 2:(1 + m)] <- beta; J[1, n] <- beta * (1 - eps)
  J[2, 1] <- sigma
  for (k in 1:m) { J[1 + k, 1 + k] <- -(r + gamma); if (k > 1) J[1 + k, k] <- r }
  J[n, 1 + m] <- r; J[n, n] <- -gamma
  J
}
growth_rate <- function(beta, sigma, gamma, tau, eps, m = 1) max(Re(eigen(jac_m(beta, sigma, gamma, tau, eps, m), only.values = TRUE)$
      values))
## Reproduction number at the disease-free state: R = (beta/gamma) * (1 - eps * a^m),  a = m*tau/(m*tau + gamma)
Re_value <- function(beta, gamma, tau, eps, m = 1) { a <- m * tau / (m * tau + gamma); (beta / gamma) * (1 - eps * a^m) }
frac_isolated <- function(gamma, tau, m = 1) (m * tau / (m * tau + gamma))^m       # share of cases isolated before recovery/death

calibrate_beta <- function(r, sigma, gamma, tau0, eps, m = 1)
  uniroot(function(b) growth_rate(b, sigma, gamma, tau0, eps, m) - r, c(1e-4, 6), tol = 1e-12)$root

make_rhs <- function(m, sigma, gamma, pop) {
  function(t, y, p) {                       # p = c(beta, tau_new, eps, tau0, ramp_days)
    beta <- p[1]; eps <- p[3]
    tau <- if (p[5] > 0) p[4] + (p[2] - p[4]) * min(1, t / p[5]) else p[2]
    S <- y[1]; E <- y[2]; I <- y[3:(2 + m)]; Q <- y[3 + m]
    lam <- beta * (sum(I) + (1 - eps) * Q) / pop
    rate <- m * tau
    dI <- c(sigma * E, rate * I[-m]) - (rate + gamma) * I
    list(c(-lam * S, lam * S - sigma * E, dI, rate * I[m] - gamma * Q, gamma * (sum(I) + Q), sigma * E))
  }
}
## Initial state: dominant eigenvector of the linearised (E, I_1..I_m, Q) system at the BASELINE delay, scaled so that sigma*E = i0
init_state <- function(beta, sigma, gamma, tau0, eps, i0, m = 1, pop = CFG$pop) {
  ev <- eigen(jac_m(beta, sigma, gamma, tau0, eps, m)); v <- abs(Re(ev$vectors[, which.max(Re(ev$values))]))
  v <- v * i0 / (sigma * v[1])
  c(S = pop - sum(v), E = v[1], I = v[2:(1 + m)], Q = v[2 + m], R = 0, C = 0)
}
## Run one scenario. Returns daily symptomatic incidence (length tend) and active infections (E+I+Q) at each day.
run_det <- function(beta, sigma, gamma, tau_new, eps, i0, tau0 = CFG$tau0, m = 1, tend = CFG$max_days,
                    ramp = 0, pop = CFG$pop, y0 = NULL) {
  if (is.null(y0)) y0 <- init_state(beta, sigma, gamma, tau0, eps, i0, m, pop)
  o <- ode(y0, 0:tend, make_rhs(m, sigma, gamma, pop), c(beta, tau_new, eps, tau0, ramp),
           method = "lsoda", rtol = 1e-7, atol = 1e-7)
  nm <- m + 5
  list(inc = diff(o[, nm + 1]), active = rowSums(o[, 3:(3 + m + 1)]) , cum = o[, nm + 1])
}
## active infections = E + sum(I_k) + Q = columns 3 .. (3+m+1) of the ode matrix (col 1 = time, col 2 = S)
outcomes <- function(sim, horizon = CFG$horizon) {
  ext <- which(sim$active < 1)[1]                       # first day with < 1 exposed/infectious person
  list(cum = sim$cum[horizon + 1] - sim$cum[1], duration = if (is.na(ext)) Inf else ext - 1,
       peak = max(sim$inc[1:horizon]), tpeak = which.max(sim$inc[1:horizon]))
}

## ------------------------------------------------------------------ discrete-time stochastic model (m = 1)
disc_matrix <- function(beta, sigma, gamma, tau, eps, dt) {
  es <- exp(-sigma * dt); et <- exp(-tau * dt); eg <- exp(-gamma * dt)
  matrix(c(es, beta * dt, beta * dt * (1 - eps), 1 - es, et * eg, 0, 0, 1 - et, eg), 3, 3, byrow = TRUE)
}
calibrate_beta_disc <- function(r, sigma, gamma, tau0, eps, dt)
  uniroot(function(b) max(Re(eigen(disc_matrix(b, sigma, gamma, tau0, eps, dt), only.values = TRUE)$values)) - exp(r * dt),
          c(1e-4, 6), tol = 1e-12)$root
init_disc <- function(beta, sigma, gamma, tau0, eps, i0, dt, pop = CFG$pop) {
  ev <- eigen(disc_matrix(beta, sigma, gamma, tau0, eps, dt)); v <- abs(Re(ev$vectors[, which.max(Re(ev$values))]))
  v * i0 * dt / ((1 - exp(-sigma * dt)) * v[1])
}
## Vectorised over replicates: every argument may be a vector of length nrep. Binomial-chain (tau-leap) transitions.
run_stoch <- function(beta, sigma, gamma, tau, eps, E0, I0, Q0, tend = CFG$max_days, dt = 0.25, pop = CFG$pop,
```

```
                    horizon = CFG$horizon) {
  n <- length(beta); S <- rep(round(pop), n); E <- round(E0); I <- round(I0); Q <- round(Q0)
  C12 <- numeric(n); ext <- rep(NA_real_, n); steps <- as.integer(tend / dt)
  pE <- 1 - exp(-sigma * dt); pQ <- 1 - exp(-tau * dt); pG <- 1 - exp(-gamma * dt)
  for (k in seq_len(steps)) {
    lam <- beta * (I + (1 - eps) * Q) / pop
    nE <- rbinom(n, S, 1 - exp(-lam * dt)); nI <- rbinom(n, E, pE)
    nQ <- rbinom(n, I, pQ); rI <- rbinom(n, I - nQ, pG); rQ <- rbinom(n, Q, pG)
    S <- S - nE; E <- E + nE - nI; I <- I + nI - nQ - rI; Q <- Q + nQ - rQ
    if (k * dt <= horizon) C12 <- C12 + nI
    done <- is.na(ext) & (E + I + Q == 0); ext[done] <- k * dt
  }
  list(cum12 = C12, ext = ext)
}

## ------------------------------------------------------------------ writing numbers for LaTeX
MACROS <- new.env()
macro_reset <- function() rm(list = ls(MACROS), envir = MACROS)
macro <- function(name, value) assign(name, as.character(value), envir = MACROS)
macro_write <- function(file) {
  nm <- sort(ls(MACROS)); lines <- vapply(nm, function(n) sprintf("\\newcommand{\\%s}{%s}", n, get(n, MACROS)), "")
  writeLines(c("% AUTO-GENERATED by R/07_figures_tables.R -- do not edit", lines), file)
}
```

R/01_data_prep.R

```
## 01_data_prep.R -- load the surveillance data, measure the reporting delay, build the two fitting datasets.
## Inputs : data/bvd_2026/*.csv  (epiforecasts/BVDOutbreakSize export, data as of 12 Sep 2026)
## Outputs: results/prep.rds, results/tables/data_summary.csv, results/positivity.csv
find_root_ <- function() { a <- commandArgs(FALSE); f <- sub("--file=", "", a[grep("--file=", a)]); normalizePath(file.path(dirname(f
      [1]), "..")) }
source(file.path(find_root_(), "R", "00_functions.R"))
set.seed(20261101)

rd <- function(f, ...) read.csv(file.path(P$bvd, f), stringsAsFactors = FALSE, ...)
wide <- rd("bvd_2026_streams_wide.csv"); wide$date <- as.Date(wide$date)
lat  <- rd("bvd_2026_onset_curve_latest.csv"); lat$onset_date <- as.Date(lat$onset_date); REPORT <- as.Date(lat$report_date[1])
vint <- rd("bvd_2026_onset_curve_all_vintages.csv", colClasses = c(sitrep = "character"))
vint$onset_date <- as.Date(vint$onset_date); vint$report_date <- as.Date(vint$report_date)
sc <- rd("bvd_2026_scalars.csv")
cb <- sc[sc$name == "confirmed_break_dates", ]
break_date  <- as.Date(gsub("[^0-9-]", "", cb$value))
break_gross <- as.numeric(sub(".*'gross_cases': \\[(\\d+)\\].*", "\\1", cb$other_fields))
stopifnot(REPORT == CFG$cutoff, length(break_date) == 1, is.finite(break_gross))

## ---- 1. reporting delay (completeness of an onset day's count after L days), from all report vintages
ref <- vint[vint$sitrep == "121" & vint$onset_date <= as.Date("2026-08-15"), c("onset_date", "confirmed_total")]
mm <- merge(vint, ref, by = "onset_date", suffixes = c("", "_ref"))
mm$lag <- as.integer(mm$report_date - mm$onset_date); mm <- mm[mm$lag >= 0 & mm$lag <= 40, ]
agg <- aggregate(cbind(confirmed_total, confirmed_total_ref) ~ lag, mm, sum)
agg$ratio <- agg$confirmed_total / agg$confirmed_total_ref
wfit <- optim(c(1.5, 10), function(p) if (any(p <= 0)) 1e9 else sum((agg$ratio - pweibull(agg$lag + 0.5, p[1], p[2]))^2))
WK <- wfit$par[1]; WS <- wfit$par[2]
delay <- list(shape = WK, scale = WS, mean = WS * gamma(1 + 1 / WK),
              q50 = qweibull(.5, WK, WS) - .5, q90 = qweibull(.9, WK, WS) - .5, n_vintages = length(unique(vint$sitrep)),
              rmse = sqrt(mean((agg$ratio - pweibull(agg$lag + 0.5, WK, WS))^2)))

## ---- 2. Fit A data: confirmed cases by symptom-onset date (latest vintage), 25 May -- 6 Sep
dA <- lat[lat$onset_date >= as.Date("2026-05-25"), c("onset_date", "confirmed_total")]
dA$lag <- as.numeric(REPORT - dA$onset_date); dA$k <- as.numeric(dA$onset_date - CFG$t0)

## ---- 3. Fit B data: confirmed increments between report dates, 2 Jun -- 12 Sep, harmonisation step removed
cum <- wide[!is.na(wide$confirmed_case_history) & wide$date >= as.Date("2026-06-02"), c("date", "confirmed_case_history")]
dB <- data.frame(date = cum$date[-1], prev = cum$date[-nrow(cum)], y = diff(cum$confirmed_case_history))
dB$ndays <- as.numeric(dB$date - dB$prev)
step <- dB$y[dB$date == break_date]; retro <- step - break_gross; dB$y[dB$date == break_date] <- break_gross
dB$u <- as.numeric(dB$date - CFG$t0); dB$u0 <- as.numeric(dB$prev - CFG$t0)
tst <- wide[!is.na(wide$tests_analysed_daily_history), c("date", "tests_analysed_daily_history")]
dB$tests <- sapply(seq_len(nrow(dB)), function(j) {
  d <- tst$tests_analysed_daily_history[tst$date > dB$prev[j] & tst$date <= dB$date[j]]
  if (length(d)) mean(d) * dB$ndays[j] else NA })
```

```r
dB$month <- format(dB$date, "%b")

## ---- 4. positivity diagnostic (new confirmed / samples analysed)
pos <- aggregate(cbind(y, tests) ~ month, dB[!is.na(dB$tests), ], sum)
pos$days <- aggregate(ndays ~ month, dB[!is.na(dB$tests), ], sum)$ndays
pos <- pos[match(c("Jun", "Jul", "Aug", "Sep"), pos$month), ]
pos$positivity <- 100 * pos$y / pos$tests; pos$tests_per_day <- pos$tests / pos$days
names(pos)[2:3] <- c("new_confirmed", "tests_analysed")
write.csv(pos, file.path(P$res, "positivity.csv"), row.names = FALSE)

## ---- 5. descriptive summary
last <- wide[wide$date == CFG$cutoff, ]
desc <- data.frame(
  quantity = c("confirmed cases (cumulative, 12 Sep)", "confirmed deaths (cumulative, 12 Sep)", "crude deaths/cases",
               "onset-curve cases (latest vintage)", "onset-curve share of confirmed", "report intervals in Fit B",
               "retrospective cases removed (22 Jul)", "mean new confirmed per day, 1-12 Sep"),
  value = c(last$confirmed_case_history, last$confirmed_death_history, last$confirmed_death_history / last$confirmed_case_history,
            sum(lat$confirmed_total), sum(lat$confirmed_total) / last$confirmed_case_history, nrow(dB), retro,
            (last$confirmed_case_history - wide$confirmed_case_history[wide$date == as.Date("2026-09-01")]) / 11))
write.csv(desc, file.path(P$tab, "data_summary.csv"), row.names = FALSE)

saveRDS(list(agg = agg, delay = delay, dA = dA, dB = dB, pos = pos, REPORT = REPORT, break_date = break_date,
             break_gross = break_gross, retro = retro, lat = lat, last = last, desc = desc), file.path(P$res, "prep.rds"))
cat(sprintf("Reporting delay: Weibull shape %.2f scale %.1f d, mean %.1f d, 50%% by %.0f d, 90%% by %.0f d (%d vintages)\n",
            WK, WS, delay$mean, delay$q50, delay$q90, delay$n_vintages))
cat(sprintf("Fit A: %d onset days | Fit B: %d report intervals | %d retrospective cases removed\n", nrow(dA), nrow(dB), retro))
print(desc); print(pos)
```

R/02_fit.R

```r
## 02_fit.R -- fit the SEIQR model to the 2026 DRC surveillance data (maximum likelihood, negative-binomial counts).
## Natural-history parameters are fixed at central literature values; estimated: starting incidence, transmission rate
## (optionally changing once on an estimated date) and count dispersion. Uncertainty: profile likelihood over the change
## date + multivariate-normal (Hessian) approximation for the other parameters.
## Outputs: results/fit.rds, results/fit_draws_{A,B}.csv, results/tables/fit_*.csv
find_root_ <- function() { a <- commandArgs(FALSE); f <- sub("--file=", "", a[grep("--file=", a)]); normalizePath(file.path(dirname(f[1]), "..")) }
source(file.path(find_root_(), "R", "00_functions.R"))
set.seed(20261101); t_start <- Sys.time()
pr <- readRDS(file.path(P$res, "prep.rds")); dA <- pr$dA; dB <- pr$dB
TEND <- as.numeric(CFG$cutoff - CFG$t0); WK <- pr$delay$shape; WS <- pr$delay$scale
cc <- function(L) pweibull(L + 0.5, WK, WS)                    # share of an onset day's cases already reported after L days
pm_full <- function(n) { pm <- diff(c(0, cc(0:(n - 1)))); M <- matrix(0, n, n); for (u in 1:n) M[u, 1:u] <- pm[u:1]; M }
PM <- pm_full(TEND)

## predictions on the observation scale
predA <- function(inc) inc[dA$k] * cc(dA$lag)
predB <- function(inc) { Cc <- c(0, cumsum(as.vector(PM %*% inc))); Cc[dB$u + 1] - Cc[dB$u0 + 1] }

## piecewise-constant transmission: beta1 until day tc, beta2 afterwards; returns daily incidence on days 1..TEND
sim_fit <- function(i0, b1, b2, tc, eps = CFG$eps_base, sigma = CFG$sigma0, gamma = CFG$gamma0, tend = TEND) {
  rhs <- make_rhs(1, sigma, gamma, CFG$pop); y0 <- init_state(b1, sigma, gamma, CFG$tau0, eps, i0, 1)
  o1 <- ode(y0, 0:min(tc, tend), rhs, c(b1, CFG$tau0, eps, CFG$tau0, 0), method = "lsoda", rtol = 1e-6, atol = 1e-6)
  C <- o1[, 7]
  if (tc < tend) {
    o2 <- ode(o1[nrow(o1), -1], tc:tend, rhs, c(b2, CFG$tau0, eps, CFG$tau0, 0), method = "lsoda", rtol = 1e-6, atol = 1e-6)
    C <- c(C, o2[-1, 7])
  }
  diff(C)
}
nll_gen <- function(p4, tc, pred, y, ...) {                    # p4 = log i0, log beta1, log beta2, log k
  inc <- tryCatch(sim_fit(exp(p4[1]), exp(p4[2]), exp(p4[3]), tc, ...), error = function(e) NULL)
  if (is.null(inc) || any(!is.finite(inc))) return(1e10)
  -sum(dnbinom(y, mu = pmax(pred(inc), 1e-8), size = exp(p4[4]), log = TRUE))
}
nm2 <- function(f, start, it = 4000) { o <- optim(start, f, control = list(maxit = it, reltol = 1e-10)); optim(o$par, f, control = list(maxit = it, reltol = 1e-12)) }
fit_const  <- function(pred, y, start, ...) { f <- function(p) nll_gen(c(p[1], p[2], p[2], p[3]), TEND, pred, y, ...); o <- nm2(f, start)
  list(par4 = c(o$par[1], o$par[2], o$par[2], o$par[3]), nll = o$value, hess = optimHess(o$par, f), tc = TEND) }
fit_change <- function(pred, y, tc, start4, ...) { f <- function(p) nll_gen(p, tc, pred, y, ...); o <- nm2(f, start4)
```

```
  list(par4 = o$par, nll = o$value, hess = optimHess(o$par, f), tc = tc) }
mvn_draw <- function(mean, hess) {
  cv <- tryCatch(solve(hess), error = function(e) diag(1 / pmax(diag(hess), 1e-6))); cv <- (cv + t(cv)) / 2
  e <- eigen(cv); as.vector(mean + e$vectors %*% (sqrt(pmax(e$values, 1e-8)) * rnorm(length(mean))))
}
gr <- function(b, eps = CFG$eps_base, sigma = CFG$sigma0, gamma = CFG$gamma0) growth_rate(b, sigma, gamma, CFG$tau0, eps, 1)

analyse <- function(label, pred, y, start_const) {
  cat("\n==== ", label, " ====\n", sep = "")
  c0 <- fit_const(pred, y, start_const)
  grid <- as.numeric(seq(as.Date("2026-06-18"), as.Date("2026-08-28"), by = 5) - CFG$t0)
  prof <- list(); for (tc in grid) prof[[as.character(tc)]] <- fit_change(pred, y, tc, c0$par4)
  best <- as.numeric(names(prof))[which.min(sapply(prof, `[[`, "nll"))]
  for (tc in setdiff((best - 4):(best + 4), grid)) prof[[as.character(tc)]] <- fit_change(pred, y, tc, prof[[as.character(best)]]$par4)
  prof <- prof[order(as.numeric(names(prof)))]
  nll_tc <- sapply(prof, `[[`, "nll"); tcs <- as.numeric(names(prof)); best <- tcs[which.min(nll_tc)]
  b <- prof[[as.character(best)]]; ok <- tcs[nll_tc - min(nll_tc) <= 1.92]
  aic0 <- 2 * c0$nll + 6; aic1 <- 2 * b$nll + 10
  cat(sprintf("constant: nll %.2f r %.4f | changepoint %s: nll %.2f r %.4f -> %.4f | dAIC %.1f | tc 95%% %s to %s\n", c0$nll,
              gr(exp(c0$par4[2])), format(CFG$t0 + best), b$nll, gr(exp(b$par4[2])), gr(exp(b$par4[3])), aic0 - aic1,
              format(CFG$t0 + min(ok)), format(CFG$t0 + max(ok))))
  list(const = c0, prof = prof, tcs = tcs, nll_tc = nll_tc, best = b, aic0 = aic0, aic1 = aic1, ok = ok)
}
A <- analyse("Fit A: onset curve (reporting-delay corrected)", predA, dA$confirmed_total, c(log(25), log(0.245), log(20)))
B <- analyse("Fit B: confirmed cases by report date", predB, dB$y, c(log(35), log(0.245), log(30)))

## ---- uncertainty draws: change date ~ profile-likelihood weights; other parameters ~ MVN(Hessian) at that change date
draws_for <- function(F, pred, fit_label, nd = 1000, nband = 200) {
  w <- exp(-(F$nll_tc - min(F$nll_tc))); w <- w / sum(w)
  out <- vector("list", nd); bands <- list()
  for (i in 1:nd) {
    tc <- sample(F$tcs, 1, prob = w); fit <- F$prof[[as.character(tc)]]
    p4 <- mvn_draw(fit$par4, fit$hess); p4[2:3] <- pmin(pmax(p4[2:3], log(0.05)), log(1.5))
    inc <- tryCatch(sim_fit(exp(p4[1]), exp(p4[2]), exp(p4[3]), tc), error = function(e) NULL)
    if (is.null(inc)) next
    out[[i]] <- data.frame(fit = fit_label, tc_day = tc, logi0 = p4[1], beta1 = exp(p4[2]), beta2 = exp(p4[3]), k = exp(p4[4]),
      r1 = gr(exp(p4[2])), r2 = gr(exp(p4[3])), Re1 = Re_value(exp(p4[2]), CFG$gamma0, CFG$tau0, CFG$eps_base),
      Re2 = Re_value(exp(p4[3]), CFG$gamma0, CFG$tau0, CFG$eps_base), inc_end = inc[TEND])
    if (length(bands) < nband) bands[[length(bands) + 1]] <- list(inc = inc, mu = pred(inc))
  }
  list(draws = do.call(rbind, out), bands = bands)
}
cat("\nSampling parameter uncertainty ...\n")
DA <- draws_for(A, predA, "A"); DB <- draws_for(B, predB, "B")
write.csv(DA$draws, file.path(P$res, "fit_draws_A.csv"), row.names = FALSE); write.csv(DB$draws, file.path(P$res, "fit_draws_B.csv"),
      row.names = FALSE)

## ---- adequacy: 95% negative-binomial predictive coverage and residual autocorrelation (at the best fit)
adequacy <- function(F, pred, y) {
  b <- F$best; inc <- sim_fit(exp(b$par4[1]), exp(b$par4[2]), exp(b$par4[3]), b$tc); mu <- pred(inc); k <- exp(b$par4[4])
  lo <- qnbinom(.025, mu = mu, size = k); hi <- qnbinom(.975, mu = mu, size = k)
  pr_res <- (y - mu) / sqrt(mu + mu^2 / k)
  c(coverage95 = mean(y >= lo & y <= hi), resid_lag1_acf = acf(pr_res, plot = FALSE, lag.max = 1)$acf[2], resid_sd = sd(pr_res), k = k)
}
adq <- rbind(A = adequacy(A, predA, dA$confirmed_total), B = adequacy(B, predB, dB$y)); print(round(adq, 3))

## ---- sensitivity of the fitted growth rate to the fixed natural-history values (Fit B, change date fixed at best)
nh <- expand.grid(latent = c(4.0, 10.1), infectious = c(5, 10)); nh_rows <- list()
for (j in 1:nrow(nh)) {
  s <- 1 / nh$latent[j]; g <- 1 / nh$infectious[j]
  f <- function(p) nll_gen(p, B$best$tc, predB, dB$y, sigma = s, gamma = g)
  o <- nm2(f, B$best$par4, 2500)
  nh_rows[[j]] <- data.frame(latent = nh$latent[j], infectious = nh$infectious[j], nll = o$value,
    beta2 = exp(o$par[3]), r_after = growth_rate(exp(o$par[3]), s, g, CFG$tau0, CFG$eps_base, 1), r_before = growth_rate(exp(o$par[2]),
       s, g, CFG$tau0, CFG$eps_base, 1),
    inc_end = sim_fit(exp(o$par[1]), exp(o$par[2]), exp(o$par[3]), B$best$tc, sigma = s, gamma = g)[TEND])
}
nhtab <- do.call(rbind, nh_rows); print(round(nhtab, 4))
write.csv(nhtab, file.path(P$tab, "fit_naturalhistory_sensitivity.csv"), row.names = FALSE)

## ---- beta at each isolation effectiveness (refit at the best change date)
eps_rows <- list()
for (e in CFG$eps) for (nm in c("A", "B")) {
```

```
  F <- if (nm == "A") A else B; pred <- if (nm == "A") predA else predB; y <- if (nm == "A") dA$confirmed_total else dB$y
  fit <- fit_change(pred, y, F$best$tc, F$best$par4, eps = e)
  eps_rows[[length(eps_rows) + 1]] <- data.frame(fit = nm, eps = e, nll = fit$nll, beta_before = exp(fit$par4[2]), beta_after = exp(fit$par4[3]),
    r_before = gr(exp(fit$par4[2]), e), r_after = gr(exp(fit$par4[3]), e),
    Re_before = Re_value(exp(fit$par4[2]), CFG$gamma0, CFG$tau0, e), Re_after = Re_value(exp(fit$par4[3]), CFG$gamma0, CFG$tau0, e))
}
epstab <- do.call(rbind, eps_rows); write.csv(epstab, file.path(P$tab, "fit_by_effectiveness.csv"), row.names = FALSE); print(round(epstab[, -1], 4))

saveRDS(list(A = A, B = B, DA = DA, DB = DB, adq = adq, nhtab = nhtab, epstab = epstab, TEND = TEND), file.path(P$res, "fit.rds"))
cat(sprintf("\nFit finished in %.1f min\n", as.numeric(difftime(Sys.time(), t_start, units = "mins"))))
```

R/03_scenarios.R

```
## 03_scenarios.R -- nine intervention scenarios (delay 7/5/3 d  x  isolation effectiveness 50/80/100%) from 12 Sep 2026.
## Uncertainty: each of the n draws carries (growth rate r, starting incidence) from the fitted posterior (Fit B, joint with the change date)
## plus latent and infectious periods drawn from triangular distributions. Beta is re-calibrated per draw and per effectiveness so that the
## 7-day baseline reproduces that draw's fitted growth rate; scenarios then change ONLY the isolation rate. All scenarios share the same draws
## (paired), so cases averted are differences within a draw.
## Also computes a continuous policy surface (delay 2--10 d x effectiveness 30--100%).
## Outputs: results/scenarios.rds, results/tables/scenario_summary.csv, results/policy_surface.csv
find_root_ <- function() { a <- commandArgs(FALSE); f <- sub("--file=", "", a[grep("--file=", a)]); normalizePath(file.path(dirname(f[1]), "..")) }
source(file.path(find_root_(), "R", "00_functions.R"))
set.seed(20261101); t_start <- Sys.time()
fit <- readRDS(file.path(P$res, "fit.rds")); D <- fit$DB$draws; n <- nrow(D)
L <- rtri(n, CFG$lat_tri); Di <- rtri(n, CFG$inf_tri); sigma <- 1 / L; gamma <- 1 / Di
i0 <- D$inc_end; r2 <- D$r2

runs <- list(); traj <- list(); k <- 0
for (eps in CFG$eps) {
  beta <- vapply(1:n, function(i) calibrate_beta(r2[i], sigma[i], gamma[i], CFG$tau0, eps, 1), 0)
  for (dl in CFG$delays) {
    tr <- matrix(NA_real_, n, CFG$horizon); out <- vector("list", n)
    for (i in 1:n) {
      s <- run_det(beta[i], sigma[i], gamma[i], 1 / dl, eps, i0[i], tend = CFG$max_days)
      o <- outcomes(s); tr[i, ] <- s$inc[1:CFG$horizon]
      out[[i]] <- data.frame(draw = i, eps = eps, delay = dl, cum12 = o$cum, duration = o$duration, peak = o$peak, tpeak = o$tpeak,
                             Re = Re_value(beta[i], gamma[i], 1 / dl, eps, 1), frac_iso = frac_isolated(gamma[i], 1 / dl, 1), beta = beta[i])
    }
    k <- k + 1; runs[[k]] <- do.call(rbind, out); traj[[sprintf("e%.1f_d%d", eps, dl)]] <- tr
    cat(sprintf("eps %.1f delay %d done (%.1f min)\n", eps, dl, as.numeric(difftime(Sys.time(), t_start, units = "mins"))))
  }
}
R <- do.call(rbind, runs)
base <- R[R$delay == 7, c("draw", "eps", "cum12")]; names(base)[3] <- "cum12_base"
R <- merge(R, base, by = c("draw", "eps")); R$averted <- R$cum12_base - R$cum12; R$pct_averted <- 100 * R$averted / R$cum12_base
R$deaths_averted <- R$averted * CFG$cfr

## ---- summary table
summ <- do.call(rbind, lapply(split(R, list(R$eps, R$delay)), function(d) {
  qa <- q3(d$averted); qp <- q3(d$pct_averted); qc <- q3(d$cum12); qr <- q3(d$Re); qd <- q3(d$deaths_averted)
  dur <- d$duration
  data.frame(eps = d$eps[1], delay = d$delay[1], cum12_med = qc[1], cum12_lo = qc[2], cum12_hi = qc[3],
    averted_med = qa[1], averted_lo = qa[2], averted_hi = qa[3], pct_med = qp[1], pct_lo = qp[2], pct_hi = qp[3],
    deaths_med = qd[1], deaths_lo = qd[2], deaths_hi = qd[3],
    Re_med = qr[1], Re_lo = qr[2], Re_hi = qr[3], p_Re_below1 = mean(d$Re < 1),
    dur_med = quantile(dur, .5, type = 1), dur_lo = quantile(dur, .025, type = 1), dur_hi = quantile(dur, .975, type = 1),
    p_end_12m = mean(dur <= 365), p_end_24m = mean(dur <= 730), p_end_60m = mean(is.finite(dur)),
    peak_med = median(d$peak), tpeak_med = median(d$tpeak), frac_iso_med = median(d$frac_iso))
}))
rownames(summ) <- NULL; summ <- summ[order(summ$eps, -summ$delay), ]
write.csv(summ, file.path(P$tab, "scenario_summary.csv"), row.names = FALSE)
print(round(summ[, c("eps", "delay", "cum12_med", "pct_med", "pct_lo", "pct_hi", "Re_med", "p_Re_below1", "p_end_12m", "p_end_24m", "p_end_60m")], 2), row.names = FALSE)
saveRDS(list(R = R, traj = traj, L = L, Di = Di, summ = summ, n = n), file.path(P$res, "scenarios.rds"))
```

```r

## ---- continuous policy surface (first 100 draws; 12-month cases)
nd <- 100; dgrid <- 2:10; egrid <- seq(0.3, 1, by = 0.1); surf <- list(); kk <- 0
for (eps in egrid) {
  beta <- vapply(1:nd, function(i) calibrate_beta(r2[i], sigma[i], gamma[i], CFG$tau0, eps, 1), 0)
  c7 <- vapply(1:nd, function(i) outcomes(run_det(beta[i], sigma[i], gamma[i], CFG$tau0, eps, i0[i], tend = CFG$horizon))$cum, 0)
  for (dl in dgrid) {
    cs <- vapply(1:nd, function(i) outcomes(run_det(beta[i], sigma[i], gamma[i], 1 / dl, eps, i0[i], tend = CFG$horizon))$cum, 0)
    re <- vapply(1:nd, function(i) Re_value(beta[i], gamma[i], 1 / dl, eps, 1), 0)
    kk <- kk + 1; surf[[kk]] <- data.frame(eps = eps, delay = dl, pct_averted = median(100 * (c7 - cs) / c7), Re = median(re), p_Re_
      below1 = mean(re < 1))
  }
}
surf <- do.call(rbind, surf); write.csv(surf, file.path(P$res, "policy_surface.csv"), row.names = FALSE)
cat(sprintf("Scenarios finished in %.1f min\n", as.numeric(difftime(Sys.time(), t_start, units = "mins"))))
```

R/04_stochastic.R

```r
## 04_stochastic.R -- stochastic (binomial-chain) version of the model: do the conclusions survive chance events?
## Same posterior draws as 03_scenarios.R (first n = 500). Transmission is calibrated in discrete time so the stochastic model has the
## same expected growth rate as the deterministic one. Outputs: results/stochastic.rds, results/tables/stochastic_summary.csv
find_root_ <- function() { a <- commandArgs(FALSE); f <- sub("--file=", "", a[grep("--file=", a)]); normalizePath(file.path(dirname(f
      [1]), "..")) }
source(file.path(find_root_(), "R", "00_functions.R"))
set.seed(20261102); t_start <- Sys.time()
sc <- readRDS(file.path(P$res, "scenarios.rds")); fit <- readRDS(file.path(P$res, "fit.rds"))
n <- 500; DT <- 0.25; D <- fit$DB$draws[1:n, ]; sigma <- 1 / sc$L[1:n]; gamma <- 1 / sc$Di[1:n]

rows <- list()
for (eps in CFG$eps) {
  beta <- vapply(1:n, function(i) calibrate_beta_disc(D$r2[i], sigma[i], gamma[i], CFG$tau0, eps, DT), 0)
  v <- t(vapply(1:n, function(i) init_disc(beta[i], sigma[i], gamma[i], CFG$tau0, eps, D$inc_end[i], DT), numeric(3)))
  for (dl in CFG$delays) {
    s <- run_stoch(beta, sigma, gamma, 1 / dl, eps, v[, 1], v[, 2], v[, 3], dt = DT)
    d <- sc$R[sc$R$draw <= n & sc$R$eps == eps & sc$R$delay == dl, ]; d <- d[order(d$draw), ]
    rows[[length(rows) + 1]] <- data.frame(eps = eps, delay = dl, draw = 1:n, cum12 = s$cum12, ext = s$ext, det_cum12 = d$cum12, det_
      dur = d$duration)
    cat(sprintf("eps %.1f delay %d done (%.1f min)\n", eps, dl, as.numeric(difftime(Sys.time(), t_start, units = "mins"))))
  }
}
S <- do.call(rbind, rows); saveRDS(S, file.path(P$res, "stochastic.rds"))
ss <- do.call(rbind, lapply(split(S, list(S$eps, S$delay)), function(d) {
  q <- q3(d$cum12); qd <- q3(d$det_cum12)
  data.frame(eps = d$eps[1], delay = d$delay[1], stoch_med = q[1], stoch_lo = q[2], stoch_hi = q[3], det_med = qd[1],
             p_end_12m = mean(d$ext <= 365, na.rm = TRUE), p_end_24m = mean(d$ext <= 730, na.rm = TRUE), p_end_60m = mean(!is.na(d$ext)
    ),
             ext_med = median(d$ext, na.rm = TRUE), det_p_end_60m = mean(is.finite(d$det_dur)))
}))
ss <- ss[order(ss$eps, -ss$delay), ]; rownames(ss) <- NULL
ss$pct_red_med <- NA; for (e in CFG$eps) { b <- ss$stoch_med[ss$eps == e & ss$delay == 7]; i <- ss$eps == e; ss$pct_red_med[i] <- 100 *
     (b - ss$stoch_med[i]) / b }
write.csv(ss, file.path(P$tab, "stochastic_summary.csv"), row.names = FALSE); print(round(ss, 2), row.names = FALSE)
cat(sprintf("Stochastic finished in %.1f min\n", as.numeric(difftime(Sys.time(), t_start, units = "mins"))))
```

R/05_sensitivity.R

```r
## 05_sensitivity.R -- (a) global sensitivity (partial rank correlation) and (b) one-at-a-time / structural variants.
## Outputs: results/tables/prcc.csv, results/tables/variants.csv
find_root_ <- function() { a <- commandArgs(FALSE); f <- sub("--file=", "", a[grep("--file=", a)]); normalizePath(file.path(dirname(f
      [1]), "..")) }
source(file.path(find_root_(), "R", "00_functions.R"))
set.seed(20261103); t_start <- Sys.time()
sc <- readRDS(file.path(P$res, "scenarios.rds")); fit <- readRDS(file.path(P$res, "fit.rds"))
DB <- fit$DB$draws; DA <- fit$DA$draws; eps_b <- CFG$eps_base

## ------------------------------------------------------------- (a) PRCC on 12-month cases
ns <- 1500; ix <- sample(nrow(DB), ns, replace = TRUE)
X <- data.frame(latent_period = rtri(ns, CFG$lat_tri), infectious_period = rtri(ns, CFG$inf_tri),
                growth_rate = DB$r2[ix], starting_incidence = DB$inc_end[ix],
```

```r
              isolation_effectiveness = runif(ns, 0.5, 1), isolation_delay = runif(ns, 3, 7))
y <- vapply(1:ns, function(i) {
  s <- 1 / X$latent_period[i]; g <- 1 / X$infectious_period[i]; e <- X$isolation_effectiveness[i]
  b <- calibrate_beta(X$growth_rate[i], s, g, CFG$tau0, e, 1)
  outcomes(run_det(b, s, g, 1 / X$isolation_delay[i], e, X$starting_incidence[i], tend = CFG$horizon))$cum }, 0)
prcc <- function(X, y) {
  Rk <- apply(X, 2, rank); yr <- rank(y)
  vapply(seq_len(ncol(X)), function(j) {
    A <- cbind(1, Rk[, -j, drop = FALSE]); rx <- Rk[, j] - A %*% qr.solve(A, Rk[, j]); ry <- yr - A %*% qr.solve(A, yr); cor(rx, ry) },
      0)
}
pe <- prcc(X, y); B <- 300
bs <- replicate(B, { i <- sample(ns, replace = TRUE); prcc(X[i, ], y[i]) })
pt <- data.frame(parameter = names(X), prcc = pe, lo = apply(bs, 1, quantile, .025), hi = apply(bs, 1, quantile, .975))
pt <- pt[order(-abs(pt$prcc)), ]; write.csv(pt, file.path(P$tab, "prcc.csv"), row.names = FALSE); print(round(pt[, -1], 3))
cat(sprintf("PRCC done (%.1f min)\n", as.numeric(difftime(Sys.time(), t_start, units = "mins"))))

## ------------------------------------------------------------------ (b) variants (isolation effectiveness 80%)
nd <- 300; idx <- 1:nd
variants <- list(
  list(name = "Reference (all uncertainty propagated)", group = "Reference"),
  list(name = "Latent period 4.0 d", group = "Natural history", L = 4.0),
  list(name = "Latent period 10.1 d", group = "Natural history", L = 10.1),
  list(name = "Infectious period 5 d", group = "Natural history", D = 5),
  list(name = "Infectious period 10 d", group = "Natural history", D = 10),
  list(name = "Growth rate at 2.5th percentile", group = "Calibration", r = quantile(DB$r2, .025)),
  list(name = "Growth rate at 97.5th percentile", group = "Calibration", r = quantile(DB$r2, .975)),
  list(name = "Starting incidence at 2.5th percentile", group = "Calibration", i0 = quantile(DB$inc_end, .025)),
  list(name = "Starting incidence at 97.5th percentile", group = "Calibration", i0 = quantile(DB$inc_end, .975)),
  list(name = "Calibrated to onset curve (Fit A) instead of report counts", group = "Calibration", src = "A"),
  list(name = "True baseline delay 4.6 d (onset-to-sample, preprint)", group = "Baseline delay", tau0 = 1 / 4.6),
  list(name = "True baseline delay 8.6 d (onset-to-result, preprint)", group = "Baseline delay", tau0 = 1 / 8.6),
  list(name = "Erlang-3 delay (less variable)", group = "Structure", m = 3),
  list(name = "Erlang-10 delay (nearly fixed delay)", group = "Structure", m = 10),
  list(name = "Delay = mean time to isolation among those isolated", group = "Structure", cond = TRUE),
  list(name = "Population at risk 10 million", group = "Structure", pop = 1e7),
  list(name = "Interventions ramp up linearly over 30 d", group = "Implementation", ramp = 30),
  list(name = "Interventions ramp up linearly over 90 d", group = "Implementation", ramp = 90))
g <- function(v, key, default = NULL) { if (key %in% names(v)) v[[key]] else default }   # exact lookup (no $ partial matching)
run_variant <- function(v) {
  src <- if (!is.null(g(v, "src"))) DA else DB
  L <- if (!is.null(g(v, "L"))) rep(v[["L"]], nd) else sc$L[idx]; Dd <- if (!is.null(g(v, "D"))) rep(v[["D"]], nd) else sc$Di[idx]
  sigma <- 1 / L; gamma <- 1 / Dd
  r <- if (!is.null(g(v, "r"))) rep(v[["r"]], nd) else src$r2[idx]; i0 <- if (!is.null(g(v, "i0"))) rep(v[["i0"]], nd) else src$inc_end
      [idx]
  m <- g(v, "m", 1); ramp <- g(v, "ramp", 0); pop <- g(v, "pop", CFG$pop)
  tau0 <- rep(g(v, "tau0", CFG$tau0), nd)
  cond <- isTRUE(g(v, "cond", FALSE))
  if (cond) tau0 <- pmax(1 / 7 - gamma, 1e-6)           # delay defined among those isolated: tau = 1/D - gamma
  tauf <- function(dl, i) if (cond) max(1 / dl - gamma[i], 1e-6) else 1 / dl
  res <- sapply(1:nd, function(i) {
    b <- calibrate_beta(r[i], sigma[i], gamma[i], tau0[i], eps_b, m)
    base <- outcomes(run_det(b, sigma[i], gamma[i], tau0[i], eps_b, i0[i], tau0 = tau0[i], m = m, tend = CFG$horizon, pop = pop))$cum
    o5 <- outcomes(run_det(b, sigma[i], gamma[i], tauf(5, i), eps_b, i0[i], tau0 = tau0[i], m = m, tend = CFG$horizon, ramp = ramp, pop
       = pop))$cum
    o3 <- outcomes(run_det(b, sigma[i], gamma[i], tauf(3, i), eps_b, i0[i], tau0 = tau0[i], m = m, tend = CFG$horizon, ramp = ramp, pop
       = pop))$cum
    c(100 * (base - o5) / base, 100 * (base - o3) / base, Re_value(b, gamma[i], tauf(3, i), eps_b, m), frac_isolated(gamma[i], tau0[i],
       m), base)
  })
  q5 <- q3(res[1, ]); q3_ <- q3(res[2, ])
  data.frame(group = v$group, variant = v$name, pct5 = q5[1], pct5_lo = q5[2], pct5_hi = q5[3], pct3 = q3_[1], pct3_lo = q3_[2], pct3_
      hi = q3_[3],
             Re3 = median(res[3, ]), frac_iso_base = median(res[4, ]), base_cases = median(res[5, ]))
}
V <- do.call(rbind, lapply(variants, function(v) { cat(" variant:", v$name, "\n"); run_variant(v) }))
write.csv(V, file.path(P$tab, "variants.csv"), row.names = FALSE); print(round(V[, -(1:2)], 1), row.names = FALSE)
cat(sprintf("Sensitivity finished in %.1f min\n", as.numeric(difftime(Sys.time(), t_start, units = "mins"))))
```

R/06_validation.R

```r
## 06_validation.R -- out-of-sample check. The model is fitted to data up to 12 Sep 2026 ONLY. We forecast confirmed cases reported up to
## 27 Sep and 28 Oct 2026 (baseline: nothing changes) and compare with later official cumulative counts (UKHSA 27 Sep; CDC 28 Oct), which
## were never used in fitting. Benchmark: constant incidence equal to the mean of the last 14 days.
## Outputs: results/validation.rds, results/tables/validation.csv
find_root_ <- function() { a <- commandArgs(FALSE); f <- sub("--file=", "", a[grep("--file=", a)]); normalizePath(file.path(dirname(f
      [1]), "..")) }
source(file.path(find_root_(), "R", "00_functions.R"))
set.seed(20261104)
pr <- readRDS(file.path(P$res, "prep.rds")); fit <- readRDS(file.path(P$res, "fit.rds")); DB <- fit$DB$draws
ext <- read.csv(file.path(P$data, "cumulative_confirmed_drc.csv"), stringsAsFactors = FALSE); ext$date <- as.Date(ext$date)
TEND <- fit$TEND; HOR <- 46; TT <- TEND + HOR; WK <- pr$delay$shape; WS <- pr$delay$scale
cc <- function(L) pweibull(L + 0.5, WK, WS); pm <- diff(c(0, cc(0:(TT - 1)))); PM <- matrix(0, TT, TT); for (u in 1:TT) PM[u, 1:u] <-
      pm[u:1]

cum0 <- pr$last$confirmed_case_history                                    # 7,200 at 12 Sep
obs <- ext[ext$date > CFG$cutoff & ext$date <= as.Date("2026-10-28"), ]
obs$u <- as.numeric(obs$date - CFG$t0); obs$observed <- obs$cum_confirmed_drc - cum0; obs$ndays <- as.numeric(obs$date - CFG$cutoff)

nd <- 500; ix <- sample(nrow(DB), nd, replace = TRUE); E <- matrix(NA, nd, HOR)           # expected reported cases per day after cutoff
sg <- CFG$sigma0; gm <- CFG$gamma0; rhs <- make_rhs(1, sg, gm, CFG$pop)
for (j in 1:nd) {
  d <- DB[ix[j], ]; y0 <- init_state(d$beta1, sg, gm, CFG$tau0, CFG$eps_base, exp(d$logi0), 1)
  p <- function(b) c(b, CFG$tau0, CFG$eps_base, CFG$tau0, 0)
  o1 <- ode(y0, 0:d$tc_day, rhs, p(d$beta1), method = "lsoda", rtol = 1e-6, atol = 1e-6); C <- o1[, 7]
  o2 <- ode(o1[nrow(o1), -1], d$tc_day:TT, rhs, p(d$beta2), method = "lsoda", rtol = 1e-6, atol = 1e-6); C <- c(C, o2[-1, 7])
  E[j, ] <- as.vector(PM %*% diff(C))[(TEND + 1):TT]
}
fc <- function(days) { cs <- rowSums(E[, 1:days, drop = FALSE]); k <- DB$k[ix]
  pred <- rnbinom(nd, mu = cs, size = k * days)                                  # observation noise: independent NB days
  list(mu = quantile(cs, c(.5, .025, .975)), pred = quantile(pred, c(.5, .025, .975))) }
wide <- read.csv(file.path(P$bvd, "bvd_2026_streams_wide.csv")); wide$date <- as.Date(wide$date)
last14 <- (cum0 - wide$confirmed_case_history[wide$date == CFG$cutoff - 14]) / 14       # mean new confirmed/day over the last 14 days
rows <- list(); for (i in seq_len(nrow(obs))) {
  f <- fc(obs$ndays[i]); nv <- last14 * obs$ndays[i]
  rows[[i]] <- data.frame(date = obs$date[i], days_ahead = obs$ndays[i], observed = obs$observed[i], model_median = f$mu[1], model_lo =
       f$mu[2], model_hi = f$mu[3],
    pred_lo = f$pred[2], pred_hi = f$pred[3], inside_param_interval = obs$observed[i] >= f$mu[2] & obs$observed[i] <= f$mu[3],
    inside_pred_interval = obs$observed[i] >= f$pred[2] & obs$observed[i] <= f$pred[3], naive = nv,
    err_model_pct = 100 * (f$mu[1] - obs$observed[i]) / obs$observed[i], err_naive_pct = 100 * (nv - obs$observed[i]) / obs$observed[i
      ],
    cum_model = cum0 + f$mu[1], cum_observed = cum0 + obs$observed[i]) }
V <- do.call(rbind, rows); write.csv(V, file.path(P$tab, "validation.csv"), row.names = FALSE); print(round(V[, -1], 1), row.names =
      FALSE)
saveRDS(list(V = V, E = E, obs = obs, cum0 = cum0), file.path(P$res, "validation.rds"))
```

tests/test_model.R

```r
## Unit tests for the model code. Run: Rscript tests/test_model.R   (stops with an error if any check fails)
source(file.path(dirname(normalizePath(sub("--file=", "", grep("--file=", commandArgs(FALSE), value = TRUE)))), "..", "R", "00_
      functions.R"))
ok <- function(cond, msg) { if (!isTRUE(cond)) stop("FAILED: ", msg); cat("pass:", msg, "\n") }
sg <- CFG$sigma0; gm <- CFG$gamma0
for (m in c(1, 3, 10)) for (eps in c(0.5, 0.8, 1)) for (tau in c(1/7, 1/3)) {
  b <- calibrate_beta(0, sg, gm, tau, eps, m)                          # beta giving zero growth
  ok(abs(Re_value(b, gm, tau, eps, m) - 1) < 1e-8, sprintf("Re = 1 exactly when growth rate = 0 (m=%d, eps=%.1f, delay=%.0f)", m, eps,
      1/tau))
}
ok(abs(frac_isolated(gm, 1/7, 1) - 0.5) < 1e-12, "exponential 7-d delay isolates 50% before removal")
ok(abs(frac_isolated(gm, 1/7, 1000) - exp(-1)) < 1e-3, "Erlang limit -> exp(-gamma*D)")
## numerical growth rate of the ODE equals the Jacobian eigenvalue
r <- 0.01; b <- calibrate_beta(r, sg, gm, CFG$tau0, 0.8, 1)
big <- 1e12                                    # no depletion: ODE growth must equal the eigenvalue
s <- run_det(b, sg, gm, CFG$tau0, 0.8, 100, tend = 200, pop = big, y0 = init_state(b, sg, gm, CFG$tau0, 0.8, 100, 1, big))
ok(abs(log(s$inc[200] / s$inc[100]) / 100 - r) < 1e-5, "ODE growth rate matches calibration target (no depletion)")
s <- run_det(b, sg, gm, CFG$tau0, 0.8, 100, tend = 200)
ok(log(s$inc[200] / s$inc[100]) / 100 < r, "susceptible depletion slows growth slightly (as it should)")
## paired scenario runs: faster isolation always reduces cases
c7 <- outcomes(run_det(b, sg, gm, 1/7, 0.8, 100, tend = 400))$cum; c3 <- outcomes(run_det(b, sg, gm, 1/3, 0.8, 100, tend = 400))$cum
ok(c3 < c7, "3-day delay gives fewer cases than 7-day")
## effectiveness 0 => delay irrelevant
b0 <- calibrate_beta(r, sg, gm, CFG$tau0, 0, 1)
```

```r
ok(abs(outcomes(run_det(b0, sg, gm, 1/3, 0, 100, tend = 400))$cum - outcomes(run_det(b0, sg, gm, 1/7, 0, 100, tend = 400))$cum) < 1e-3 * c7, "eps = 0 makes delay irrelevant")
## stochastic mean ~ deterministic
set.seed(1); n <- 400; dt <- 0.25
bd <- calibrate_beta_disc(0.01, sg, gm, CFG$tau0, 0.8, dt); v <- init_disc(bd, sg, gm, CFG$tau0, 0.8, 100, dt)
st <- run_stoch(rep(bd, n), sg, gm, 1/7, 0.8, rep(v[1], n), rep(v[2], n), rep(v[3], n), tend = 365)
dt_cum <- outcomes(run_det(b, sg, gm, 1/7, 0.8, 100, tend = 400))$cum
ok(abs(mean(st$cum12) / dt_cum - 1) < 0.05, sprintf("stochastic mean (%.0f) within 5%% of deterministic (%.0f)", mean(st$cum12), dt_cum))
cat("ALL TESTS PASSED\n")
```